Beam Structured Precoder for HF Skywave Massive MIMO-OFDM Communications With Channel Smoothness Constraint

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***Abstract*—In this paper, we investigate precoder design for high frequency (HF) skywave massive multiple-input multiple-output (MIMO) communications with orthogonal frequency division multiplexing (OFDM) modulation. We first reveal the effect of the precoder on the effective channel at receivers and formulate the precoder design for a group of subcarriers as a sum-rate maximization problem, where the delay spread of the effective channel is constrained to maintain its smoothness. Then with the beam based channel model and beam domain channel sparsity, the design of space domain precoders for a group of subcarriers are transformed into that of a space-frequency (SF) beam domain vector and the resulting space domain precoder at each subcarrier is beam structured. Efficient calculation for design and implementation of the beam structured precoder (BSP) is proposed. Moreover, effective channel estimation with the BSP is discussed. Simulation results show that the proposed BSP can enhance the effective channel estimation performance and significantly improve the system performance.**

***Index Terms*—Massive MIMO-OFDM, HF skywave communications, beam structured precoder, channel estimation.**

I. Introduction

FUTURE wireless network is required to provide continuous global service coverage, including remote and rural areas [1], [2]. High frequency (HF) communications, operating in the 3 to 30 MHz frequency range, can achieve long-range over-the-horizon transmission via ionospheric reflection and provide worldwide coverage [3], [4]. However, due to the challenging ionospheric conditions and limited spectrum resources, traditional point-to-point HF communications suffer from low data rates and limited usability.

By deploying a large number of antennas at the base station (BS) to serve multiple user terminals (UTs) in the same time-frequency resource simultaneously, massive multiple-input multiple-output (MIMO) technology can significantly boost spectral efficiency [5], [6], [7], [8], [9]. Recently, massive MIMO in combination with orthogonal frequency division multiplexing (OFDM) has been integrated into HF skywave communications and significantly improved system performance [10], [11], [12], [13]. In [10], the asymptotically achievable sum-rates of minimum mean-squared error (MMSE) based uplink (UL) receiver and downlink (DL) precoder are derived to show the significant performance gain for HF communications. In [11], a triple-beam based channel model is proposed and the low-complexity channel estimation algorithm is given. In [12] and [13], low-complexity signal detectors are proposed by exploiting the beam domain channel sparsity of HF skywave massive MIMO communications.

The precoder design is crucial for DL massive MIMO transmission to suppress the multi-user interference and improve the sum-rate performance and has been investigated in terrestrial cellular massive MIMO systems. Although nonlinear precoders, such as dirty-paper coding (DPC) [14], can achieve optimal performance, their complexity is typically unbearable. Therefore, linear precoders have been widely investigated and used in massive MIMO systems [15], such as maximal ratio transmission (MRT) [16], zero-forcing (ZF) [17], regularized ZF (RZF) [18], [19], [20], MMSE [21] and signal to leakage noise ratio (SLNR) precoder [22]. Moreover, weighted sum-rate (WSR) is also a criterion with great practical significance, according to which the weighted MMSE (WMMSE) precoder [23], [24] and matrix manifold optimization based precoder are designed [25]. The precoder design is also investigated in HF skywave massive MIMO communications [26], [27]. In massive MIMO-OFDM systems, the aforementioned precoders can be applied individual at each subcarrier, which, however, may make the effective channel of each UT (i.e., the concatenation of the precoders and the channel of each UT) unsmooth in the frequency domain. In other words, the

Received 5 December 2024; revised 14 April 2025; accepted 29 June 2025. Date of publication 14 July 2025; date of current version 14 November 2025. This work was supported by the Jiangsu Province Basic Research Project under Grants BK20192002 and BK20212001, the Jiangsu Province Major Science and Technology Project under Grant BG2024005, the Fundamental Research Funds for the Central Universities under Grants 2242022k60007 and 2242022K60002, the Key R&D Plan of Jiangsu Province under Grant BE2022067, the Civil Aerospace Technology Research Project under Grant D040304, the National Natural Science Foundation of China under Grants 62331024 and U22B2006, and the Huawei Cooperation Project. The associate editor coordinating the review of this article and approving it for publication was N. Lee. *(Corresponding author: Xiqi Gao.)*

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Digital Object Identifier 10.1109/TCOMM.2025.3588564

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effective channel correlation between adjacent subcarriers is weakened, and the effective channel varies faster along the subcarriers, which complicates the effective channel estimation at receivers and ultimately leads to the system performance degradation.

To smooth the effective channel and facilitate the receiver channel estimation, the precoders for a group of subcarriers should be designed jointly. Similar problem has been considered and investigated in the wireless local area network (WLAN) [28], [29], [30], [31]. In [28], by exploiting the high correlation between adjacent subcarriers, the smoothed singular value decomposition (SVD) algorithm is proposed, based on which a frequency smoothed beamformer is proposed to generate smooth effective channels across a group of subcarriers. In [29], a smooth beamforming technique is proposed by incrementally computing the beamforming weights across subcarriers to ensure the smoothness of the effective channel. In [30], a joint transceiver design is proposed to provide a smooth beam-steering matrix and enable channel smoothing. In [31], smoothed beamforming vectors are designed by some optimization techniques to preserve the smoothness of the beamformed channel across successive subcarriers. In [32], a phase-rotated SVD based precoder is proposed by exploiting the phase non-uniqueness of SVD to smooth the effective channel and facilitate the channel estimation. However, the above works focus on the single-user scenario while the inter-user interference should be considered in massive MIMO-OFDM systems.

In this paper, we investigate the precoder design for HF skywave massive MIMO-OFDM communications while the smoothness of the effective channel is considered to facilitate the channel estimation and thus improve the system performance. The main contributions are summarized as follows.

- With beam based channel models, we reveal the relationship between the effective channels in the frequency domain and the delay domain (i.e., frequency beam domain) and show how the precoder affects the smoothness of the frequency domain effective channel.
- We design space domain precoders for a group of subcarriers jointly and formulate it as a sum-rate maximization problem with total power constraint, where the delay spread of the effective channel is also constrained to maintain its smoothness. Since the direct design of space domain precoders for a group of subcarriers has high complexity, we transform it into that of a space-frequency (SF) beam domain vector. The resulting space domain precoder at each subcarrier is beam structured. Furthermore, the used space beams in the beam structured precoder (BSP) design are related to those that each UT occupies and a space beam selection criterion for BSP is provided.
- By solving the SF beam domain vector optimization problem, the BSP design is proposed and the space domain precoder at each subcarrier can then be obtained with the beam matrices and the SF beam domain vector. With the beam structure of the precoder, by exploiting the specific structure of beam matrices and the asymptotic orthogonality among space beams, efficient calculation for the design and implementation of the BSP is presented.
- The DL effective channel estimation is discussed when the proposed BSP is used. Simulation results show that the proposed BSP can significantly enhance effective channel estimation and signal detection performances.

The rest of the paper is organized as follows. In Section II, we present the system model for HF skywave massive MIMO-OFDM communications. In Section III, the BSP is proposed, where the effective channel smoothness constraint is considered and the efficient calculation for design and implementation is given. In Section IV, the effective channel estimation is discussed. Simulation results are provided in Section V and the paper is concluded in Section VI.

Notations: The uppercase (lowercase) boldface letters denote matrices (vectors). The superscripts $(\cdot)^*$, $(\cdot)^{\mathrm{T}}$ and $(\cdot)^{\mathrm{H}}$ denote the conjugate, transpose and conjugated-transpose, respectively. $\bar{\jmath} = \sqrt{-1}$. $\|\mathbf{x}\|$ denotes the ℓ_2-norm of $\mathbf{x}$. $|\Psi|$ denotes the cardinality of set Ψ. $\Psi_1 \times \Psi_2$ denotes the Cartesian product of sets Ψ_1 and Ψ_2. $\lceil x \rceil$ denotes the smallest integer that is not less than x while $\lfloor x \rfloor$ denotes the largest integer that is not greater than x. $\circ$ and $\otimes$ denote the Hadamard and the Kronecker product operators, respectively. $[\mathbf{x}]_a$ and $[\mathbf{x}]_{\mathcal{S}}$ denote the a-th element and elements with index set $\mathcal{S}$ of vector $\mathbf{x}$, respectively. $[\mathbf{X}]_{a,b}$, $[\mathbf{X}]_{:,b}$, $[\mathbf{X}]_{a:b,c:d}$, and $[\mathbf{X}]_{:,\mathcal{S}}$ denote the (a,b)-th element, the b-th colomn, rows a to b and columns c to d, and columns with index set $\mathcal{S}$ of $\mathbf{X}$, respectively. $[\mathcal{S}]_a$ denotes the a-th element of an ordered set $\mathcal{S}$. $\mathbf{I}_N$ denotes the identity matrix of dimension N. $\mathbf{0}$ denote all-zero vector or matrix. $\mathrm{diag}\{\mathbf{x}\}$ denotes the diagonal matrix with $\mathbf{x}$ along its diagonal, and $\mathrm{diag}\{\mathbf{X}_1, \cdots, \mathbf{X}_N\}$ is the block-diagonal matrix with matrices $\mathbf{X}_1, \cdots, \mathbf{X}_N$ along the diagonals. For a matrix $\mathbf{X} \in \mathbb{C}^{M \times N}$, $\mathrm{blkcirc}\{\mathbf{X}\} = \left[\mathbf{X}, \left[[\mathbf{X}]_{M-1,:}^{\mathrm{T}} [\mathbf{X}]_{0:M-2,:}^{\mathrm{T}}\right]^{\mathrm{T}}, \cdots, \left[[\mathbf{X}]_{1:M-1,:}^{\mathrm{T}} [\mathbf{X}]_{0,:}^{\mathrm{T}}\right]^{\mathrm{T}}\right] \in \mathbb{C}^{M \times MN}$ denotes a block-circulant matrix.

II. System Model

In this section, we present the system configuration and beam based channel model for HF skywave massive MIMO-OFDM communications.

A. System Configuration

We consider a long-range HF skywave massive MIMO-OFDM system. A uniform linear array of M antennas[1] is deployed at the BS to provide long-range communications for U single-antenna UTs. The index set of UTs is $\mathcal{U} = \{0, 1, \cdots, U-1\}$. Since the carrier frequency, f_{c}, needs to be adjusted according to the different ionospheric conditions in HF skywave communications, we set inter-antenna spacing according to the highest used carrier frequency f_{o}, that is, $d = \lambda_{\mathrm{o}}/2$, where $\lambda_{\mathrm{o}} = c/f_{\mathrm{o}}$ is the wavelength, and c is the speed of light. Note that the proposed precoder with channel smoothness constraint can

[1]Note that uniform linear array is a reasonable configuration in the HF band and a typical example is the HF skywave over-the-horizon radar, where arrays deploy hundreds of antennas, leading to typically large antenna array aperture [33], [34], [35].

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be also applied to generic massive MIMO communications with different carrier frequencies and different system configurations although we focus on the HF skywave massive MIMO communications in this paper.

The number of subcarriers is denoted by $N_{\rm c}$, where $N_{\rm v}$ valid subcarriers are used for data transmission with the index set $\mathcal{K}=\{k_0,k_1,\cdots,k_{N_{\rm v}-1}\}$. The length of the cyclic prefix (CP) and the subcarrier spacing are denoted by $N_{\rm g}$ and Δ_f, respectively. The sampling interval length is $T_{\rm s}=1/(N_{\rm c}\Delta_f)$ and the duration of each OFDM symbol with CP is $T_{\rm sym}=(N_{\rm c}+N_{\rm g})\,T_{\rm s}$.

B. Beam Based Channel Model

We assume that the channel state keeps constant within an OFDM symbol and varies symbol by symbol due to the Doppler effect. During the DL transmission, the received signal at the k-th subcarrier on the n-th OFDM symbol at UT u is

$$y_{u,n,k}=\mathbf{h}_{u,n,k}^{\rm H}\sum_{u'\in\mathcal{U}}\mathbf{x}_{u',n,k}+z_{u,n,k}, \tag{1}$$

where $\mathbf{x}_{u,n,k}\in\mathbb{C}^M$ is the transmitted signal for UT u, $z_{u,n,k}$ is the additive white Gaussian noise with distribution $\mathcal{CN}\big(0,\sigma_u^2\big)$, and $\mathbf{h}_{u,n,k}^{\rm H}$ is the DL space domain channel of UT u at the k-th subcarrier on the n-th OFDM symbol, which is

$$\mathbf{h}_{u,n,k}^{\rm H}=\sum_{p=0}^{P_u-1}\gamma_{u,p}e^{\bar{\jmath}2\pi\nu_{u,p}nT_{\rm sym}}e^{-\bar{\jmath}2\pi k\Delta_f\tau_{u,p}}\mathbf{v}^{\rm H}\left(\Omega_{u,p}\right), \tag{2}$$

where

$$\mathbf{v}\left(\Omega\right)=\left[1,e^{-\bar{\jmath}2\pi f_{\rm c}\Delta_\tau\Omega},\cdots,e^{-\bar{\jmath}2\pi f_{\rm c}(M-1)\Delta_\tau\Omega}\right]\in\mathbb{C}^M \tag{3}$$

is the steering vector in the space domain. In (2), P_u is the number of paths between the BS and UT u, $\gamma_{u,p}=\beta_{u,p}e^{\bar{\jmath}\varphi_{u,p}}$ is the complex-valued gain of the p-th path of UT u, where $\beta_{u,p}$ is the gain and $\varphi_{u,p}$ is the initial phase uniformly distributed over $[0,2\pi)$. $\nu_{u,p}$ is the Doppler frequency caused by the ionosphere and the UT mobility. $\Omega_{u,p}\triangleq\sin\theta_{u,p}^{\rm az}\cos\theta_{u,p}^{\rm el}$ is the directional cosine, where $\theta_{u,p}^{\rm az}$ and $\theta_{u,p}^{\rm el}$ are the azimuth angle of arrival (AoA) and elevation AoA, respectively. $\Delta_\tau=d/c$ and $\tau_{u,p}$ is the p-th propagation path delay between UT u and the BS. The DL space-frequency domain channel is given by stacking the space domain channels at a group of subcarriers as

$$\begin{aligned}\mathbf{H}_{u,n}^{\rm H}&=\left[\mathbf{h}_{u,n,k_0},\mathbf{h}_{u,n,k_1},\cdots,\mathbf{h}_{u,n,k_{N_{\rm v}-1}}\right]^{\rm H}\\&=\sum_{p=0}^{P_u-1}\gamma_{u,p}e^{\bar{\jmath}2\pi\nu_{u,p}nT_{\rm sym}}\mathbf{u}\left(\tau_{u,p}\right)\mathbf{v}^{\rm H}\left(\Omega_{u,p}\right),\end{aligned} \tag{4}$$

where

$$\mathbf{u}\left(\tau\right)=\left[e^{-\bar{\jmath}2\pi k_0\Delta_f\tau},e^{-\bar{\jmath}2\pi k_1\Delta_f\tau},\cdots,e^{-\bar{\jmath}2\pi k_{N_{\rm v}-1}\Delta_f\tau}\right]\in\mathbb{C}^{N_{\rm v}} \tag{5}$$

is the steering vector in the frequency domain.

Note that path parameters $\Omega_{u,p}$ and $\tau_{u,p}$ are limited within sets $[-1,1)$ and $[0,\tau_{\max})$, respectively, where $\tau_{\max}=N_{\rm g}T_{\rm s}$ is the maximum delay spread. We uniformly divide these two sets into multiple disjoint subsets as $\Lambda_{n_{\rm an}}^{\rm an}=\left[\Omega_{n_{\rm an}},\Omega_{n_{\rm an}}+2/N_{\rm an}\right),n_{\rm an}\in\{0,\cdots,N_{\rm an}-1\}$ and $\Lambda_{n_{\rm de}}^{\rm de}=\left[\tau_{n_{\rm de}},\tau_{n_{\rm de}}+N_\tau/(N_{\rm de}N_{\rm v}\Delta_f)\right),n_{\rm de}\in\{0,\cdots,N_{\rm de}-1\}$, respectively, where $N_\tau=N_{\rm v}N_{\rm g}/N_{\rm c}$, $\Omega_{n_{\rm an}}=(2n_{\rm an}-N_{\rm an})/N_{\rm an}$, and $\tau_{n_{\rm de}}=N_\tau n_{\rm de}/(N_{\rm de}N_{\rm v}\Delta_f)$. The numbers of the divided subsets for the directional cosines and the delays are $N_{\rm an}=\lceil F_{\rm an}Mf_{\rm c}/f_{\rm o}\rceil$ and $N_{\rm de}=F_{\rm de}N_\tau$, respectively, where $F_{\rm an}$ and $F_{\rm de}$ are referred to as fine factors.[2] Denote $\Gamma_u^{\rm S}=\{\Omega_{u,0},\Omega_{u,1},\cdots,\Omega_{u,P_u-1}\}$, $\Gamma_u^{\rm SF}=\left\{\left(\Omega_{u,0},\tau_{u,0}\right),\left(\Omega_{u,1},\tau_{u,1}\right),\cdots,\left(\Omega_{u,P_u-1},\tau_{u,P_u-1}\right)\right\}$, and $\Lambda_{n_{\rm an},n_{\rm de}}=\Lambda_{n_{\rm an}}^{\rm an}\times\Lambda_{n_{\rm de}}^{\rm de}$, the space domain channel in (2) and the space-frequency domain channel in (4) can be respectively approximated by [11]

$$\mathbf{h}_{u,n,k}^{\rm H}\approx\tilde{\mathbf{h}}_{u,n,k}^{\rm H}\mathbf{V}^{\rm H}, \tag{6}$$

$$\mathbf{H}_{u,n}^{\rm H}\approx\mathbf{U}\tilde{\mathbf{H}}_{u,n}^{\rm H}\mathbf{V}^{\rm H}, \tag{7}$$

where $\tilde{\mathbf{h}}_{u,n,k}^{\rm H}$ is the space beam domain channel with

$$\left[\tilde{\mathbf{h}}_{u,n,k}^{\rm H}\right]_{n_{\rm an}}=\sqrt{M}\sum_{\Omega_{n_{\rm an}}\in\Gamma_u^{\rm S}\cap\Lambda_{n_{\rm an}}^{\rm an}}\gamma_{u,p}e^{\bar{\jmath}2\pi\nu_{u,p}nT_{\rm sym}}e^{-\bar{\jmath}2\pi k\Delta_f\tau_{u,p}}, \tag{8}$$

$\tilde{\mathbf{H}}_{u,n}^{\rm H}$ is the SF beam domain channel with

$$\left[\tilde{\mathbf{H}}_{u,n}^{\rm H}\right]_{n_{\rm de},n_{\rm an}}=\sqrt{MN_{\rm v}}\sum_{\left(\Omega_{n_{\rm an}},\tau_{n_{\rm de}}\right)\in\Gamma_u^{\rm SF}\cap\Lambda_{n_{\rm an},n_{\rm de}}}\gamma_{u,p}e^{\bar{\jmath}2\pi\nu_{u,p}nT_{\rm sym}}, \tag{9}$$

and $\mathbf{V}=\frac{1}{\sqrt{M}}\left[\mathbf{v}(\Omega_0),\mathbf{v}(\Omega_1),\cdots,\mathbf{v}(\Omega_{N_{\rm an}-1})\right]\in\mathbb{C}^{M\times N_{\rm an}}$ is the space beam matrix, whose columns are referred to as the space beam while $\mathbf{U}=\frac{1}{\sqrt{N_{\rm v}}}\left[\mathbf{u}\left(\tau_0\right),\mathbf{u}\left(\tau_1\right),\cdots,\mathbf{u}\left(\tau_{N_{\rm de}-1}\right)\right]\in\mathbb{C}^{N_{\rm v}\times N_{\rm de}}$ is the frequency beam matrix, whose columns are referred to as the frequency beam. Thus (6) and (7) are referred to as the beam based channel models.

III. Beam Structured Precoder

In this section, we investigate the precoder design for a group of subcarriers jointly. We formulate the precoder design as a sum-rate maximization problem, where the channel smoothness constraint is introduced. Moreover, the design of the space domain precoders for a group of subcarriers is transformed into that of an SF beam domain vector. The resulting space domain precoder at each subcarrier can be obtained from the SF beam domain vector and is beam structured. Efficient calculation for design and implementation of BSP is also proposed.

A. Problem Formulation

We consider the DL transmission on the n-th OFDM symbol and omit index n in the subscript hereafter for brevity. Recalling (1), the received signal at the k-th subcarrier at UT u can be rewritten as

$$y_{u,k}=\mathbf{h}_{u,k}^{\rm H}\sum_{u'\in\mathcal{U}}\mathbf{p}_{u',k}x_{u',k}+z_{u,k}, \tag{10}$$

[2]The sampling interval for the channel parameters can be flexibly adjusted by fine factors to reduce the approximation error caused by the grid mismatch, which is different from the conventional discrete Fourier transform (DFT) based channel model in [36].

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where $\mathbf{p}_{u,k} \in \mathbb{C}^M$ is the space domain precoder of UT u at the k-th subcarrier, $x_{u,k}$ is the data symbol intended for UT u with zero mean and unit variance. We assume that the perfect channel state information (CSI) is known at the BS.[3] Then the rate of UT u at the k-th subcarrier is given by

$$\mathcal{R}_{u,k} = \log\left(1 + \frac{\mathbf{h}_{u,k}^{\mathrm{H}}\mathbf{p}_{u,k}\mathbf{p}_{u,k}^{\mathrm{H}}\mathbf{h}_{u,k}}{\sigma_u^2 + \sum\limits_{u' \in \mathcal{U}\setminus u} \mathbf{h}_{u,k}^{\mathrm{H}}\mathbf{p}_{u',k}\mathbf{p}_{u',k}^{\mathrm{H}}\mathbf{h}_{u,k}}\right), \quad (11)$$

and the conventional per subcarrier precoder design in terms of the weighted sum-rate maximization with total power constraint is

$$\begin{aligned}\left\{\mathbf{p}_{u,k}^{\mathrm{op}}\right\}_{u\in\mathcal{U}} = &\arg\max_{\{\mathbf{p}_{u,k}\}_{u\in\mathcal{U}}} \sum_{u\in\mathcal{U}} w_u \mathcal{R}_{u,k} \\ &\text{s.t.} \sum_{u\in\mathcal{U}} \mathbf{p}_{u,k}^{\mathrm{H}}\mathbf{p}_{u,k} \le P_k, \end{aligned} \quad (12)$$

where w_u is the priority weighted factor of UT u and P_k denotes the power budget at the k-th subcarrier.

By stacking the received signal at a group of subcarriers for UT u, we have

$$\begin{aligned}\mathbf{y}_u &= \mathbf{X}_u\bar{\mathbf{H}}_u^{\mathrm{H}}\mathbf{p}_u + \sum_{u'\in\mathcal{U}\setminus u} \mathbf{X}_{u'}\bar{\mathbf{H}}_u^{\mathrm{H}}\mathbf{p}_{u'} + \mathbf{z}_u \\ &= \mathbf{X}_u\mathbf{h}_{u,u}^{\mathrm{e}} + \sum_{u'\in\mathcal{U}\setminus u} \mathbf{X}_{u'}\mathbf{h}_{u,u'}^{\mathrm{e}} + \mathbf{z}_u, \end{aligned} \quad (13)$$

where

$$\begin{aligned}\mathbf{y}_u &= \left[y_{u,k_0}, y_{u,k_1}, \cdots, y_{u,k_{N_{\mathrm{v}}-1}}\right]^{\mathrm{T}} \in \mathbb{C}^{N_{\mathrm{v}}}, \\ \mathbf{X}_u &= \mathrm{diag}\left\{x_{u,k_0}, x_{u,k_1}, \cdots, x_{u,k_{N_{\mathrm{v}}-1}}\right\} \in \mathbb{C}^{N_{\mathrm{v}}\times N_{\mathrm{v}}}, \\ \bar{\mathbf{H}}_u^{\mathrm{H}} &= \mathrm{diag}\left\{\mathbf{h}_{u,k_0}^{\mathrm{H}}, \mathbf{h}_{u,k_1}^{\mathrm{H}}, \cdots, \mathbf{h}_{u,k_{N_{\mathrm{v}}-1}}^{\mathrm{H}}\right\} \in \mathbb{C}^{N_{\mathrm{v}}\times MN_{\mathrm{v}}}, \\ \mathbf{p}_u &= \left[\mathbf{p}_{u,k_0}^{\mathrm{T}}, \mathbf{p}_{u,k_1}^{\mathrm{T}}, \cdots, \mathbf{p}_{u,k_{N_{\mathrm{v}}-1}}^{\mathrm{T}}\right]^{\mathrm{T}} \in \mathbb{C}^{MN_{\mathrm{v}}}, \\ \mathbf{z}_u &= \left[z_{u,k_0}, z_{u,k_1}, \cdots, z_{u,k_{N_{\mathrm{v}}-1}}\right]^{\mathrm{T}} \in \mathbb{C}^{N_{\mathrm{v}}}, \end{aligned}$$

and

$$\mathbf{h}_{u,u'}^{\mathrm{e}} = \bar{\mathbf{H}}_u^{\mathrm{H}}\mathbf{p}_{u'} \in \mathbb{C}^{N_{\mathrm{v}}} \quad (14)$$

is the effective channel between the intended signal for UT u' and the received signal at UT u. Note that the conventional precoder design in (12) is performed at each subcarrier individually and may make the frequency domain effective channel, $\mathbf{h}_{u,u}^{\mathrm{e}}$, unsmooth, which means that the channel correlation between the adjacent subcarriers may be weakened and the channel may vary faster along the subcarriers. Thus it will make the effective channel estimation more difficult, which is not conducive to the receiver design of UT u. To facilitate the effective channel estimation at each UT and ultimately improve the system performance, we propose to jointly design precoders for a group of subcarriers.

Similar to frequency beam matrix $\mathbf{U}$, we denote $\bar{\mathbf{U}} = \frac{1}{\sqrt{\bar{N}_{\mathrm{de}}}}\left[\mathbf{u}\left(\tau_0\right), \mathbf{u}\left(\tau_1\right), \cdots, \mathbf{u}\left(\tau_{\bar{N}_{\mathrm{de}}-1}\right)\right] \in \mathbb{C}^{N_{\mathrm{v}}\times\bar{N}_{\mathrm{de}}}$, $\bar{N}_{\mathrm{de}} = F_{\mathrm{de}}N_{\mathrm{v}}$, which can be regarded as an extended version of $\mathbf{U}$. Then similar to the beam based channel model in (7), the frequency domain effective channel, $\mathbf{h}_{u,u}^{\mathrm{e}}$, can be expressed as

$$\mathbf{h}_{u,u}^{\mathrm{e}} = \bar{\mathbf{U}}\tilde{\mathbf{h}}_{u,u}^{\mathrm{e}}, \quad (15)$$

where $\tilde{\mathbf{h}}_{u,u}^{\mathrm{e}} \in \mathbb{C}^{\bar{N}_{\mathrm{de}}}$ is referred to as the delay domain effective channel. The channel smoothness actually depends on the delay domain effective channel, $\tilde{\mathbf{h}}_{u,u}^{\mathrm{e}}$. Specifically, the larger delay spread of delay domain effective channel, the lower correlation of the frequency domain effective channels across subcarriers, and the faster the effective channel varies along the subcarriers, which reduces the channel smoothness. Therefore, we propose to limit the delay spread of $\tilde{\mathbf{h}}_{u,u}^{\mathrm{e}}$ to maintain the channel smoothness and facilitate the DL effective channel estimation at each UT. We denote N_{e} as the constrained delay spread of $\tilde{\mathbf{h}}_{u,u}^{\mathrm{e}}$, that is, only the first N_{e} elements of $\tilde{\mathbf{h}}_{u,u}^{\mathrm{e}}$ can be non-zero and the rest are zero, which can be expressed as

$$\left[\tilde{\mathbf{h}}_{u,u}^{\mathrm{e}}\right]_{N_{\mathrm{e}}:\bar{N}_{\mathrm{de}}-1} = \mathbf{0}, \quad (16)$$

We refer to (16) as the channel smoothness constraint. Then the precoder design for a group of subcarriers are formulated as the following problem

$$\begin{aligned}\left\{\mathbf{p}_u^{\mathrm{op}}\right\}_{u\in\mathcal{U}} = &\arg\max_{\{\mathbf{p}_u\}_{u\in\mathcal{U}}} \sum_{k\in\mathcal{K}}\sum_{u\in\mathcal{U}} w_u\mathcal{R}_{u,k} \\ &\text{s.t.} \sum_{u\in\mathcal{U}} \mathbf{p}_u^{\mathrm{H}}\mathbf{p}_u \le P \\ &\left[\tilde{\mathbf{h}}_{u,u}^{\mathrm{e}}\right]_{N_{\mathrm{e}}:\bar{N}_{\mathrm{de}}-1} = \mathbf{0}, \forall u \in \mathcal{U}, \end{aligned} \quad (17)$$

where $P = \sum_{k\in\mathcal{K}} P_k$ is the power budget for a group of subcarriers. Note that the problem formulated in (17) is not simply a generalization of the classic narrowband precoder design problem since not only the total power constraint but also the channel smoothness constraint are considered, thus existing techniques, such as [37], cannot be directly applied to solve such a problem. So far, the relationship between the delay domain effective channel and the precoder is not clear and the problem in (17) is intractable.

B. Precoder Design

To reveal the effect of the precoder on the effective channel, we have the following lemma, proved in Appendix A.

Lemma 1: Let the stacked precoders for a group of subcarriers, $\mathbf{p}_u$, has the following structure

$$\mathbf{p}_u = \left(\bar{\mathbf{U}}_u \otimes \mathbf{I}\right)\hat{\mathbf{p}}_u, \quad (18)$$

where $\bar{\mathbf{U}}_u = \left[\bar{\mathbf{U}}\right]_{:,0:\bar{N}_{\mathrm{de},u}-1}$, $\bar{N}_{\mathrm{de},u}$ is the number of used frequency beams, and $\hat{\mathbf{p}}_u \in \mathbb{C}^{M\bar{N}_{\mathrm{de},u}}$. Then the delay domain effective channel, $\tilde{\mathbf{h}}_{u,u}^{\mathrm{e}}$, can be given by

$$\left[\tilde{\mathbf{h}}_{u,u}^{\mathrm{e}}\right]_i = \begin{cases}\left[\hat{\mathbf{H}}_u^{\mathrm{tru}}\hat{\mathbf{p}}_u\right]_i, & 0 \le i \le \bar{N}_{\mathrm{e},u}, \\ 0, & \text{otherwise}, \end{cases} \quad (19)$$

where $\bar{N}_{\mathrm{e},u} = \min\left(N_{\mathrm{de}} + \bar{N}_{\mathrm{de},u} - 1, \bar{N}_{\mathrm{de}}\right)$, and

$$\hat{\mathbf{H}}_u^{\mathrm{tru}} = \left[\mathrm{blkcirc}\left\{\hat{\mathbf{H}}_u\right\}\right]_{0:\bar{N}_{\mathrm{e},u}-1,0:M\bar{N}_{\mathrm{de},u}-1} \quad (20)$$

is a truncated version of $\mathrm{blkcirc}\left\{\hat{\mathbf{H}}_u\right\}$, and $\hat{\mathbf{H}}_u = \frac{1}{\sqrt{N_{\mathrm{v}}}}\tilde{\mathbf{H}}_u^{\mathrm{H}}\mathbf{V}^{\mathrm{H}} \in \mathbb{C}^{N_{\mathrm{de}}\times M}$.

[3]The CSI can be acquired by the channel estimation and the channel prediction approach proposed in [11].

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Lemma 1 provides the relationship between the delay domain effective channel and the precoder. Specifically, $\tilde{\mathbf{h}}^{\mathrm{e}}_{u,u}$ can be prescribed by carefully designing $\hat{\mathbf{p}}_u$. Moreover, from Lemma 1, the delay spread of $\tilde{\mathbf{h}}^{\mathrm{e}}_{u,u}$ can be constrained to be no greater than $\bar{N}_{\mathrm{e},u}$, which depends on the number of used frequency beams, $\bar{N}_{\mathrm{de},u}$. When $N_{\mathrm{e}} = \bar{N}_{\mathrm{e},u}$, the channel smoothness constraint in (16) can be naturally satisfied with the structure in (18). Besides, the spatial dimensions introduced by massive MIMO allow the actual delay spread of the effective channel to be further reduced below $\bar{N}_{\mathrm{e},u}$. Thus when $N_{\mathrm{e}} < \bar{N}_{\mathrm{e},u}$, the channel smoothness constraint in (16) can be transformed into

$$\mathbf{E}_{N_{\mathrm{e}}}\hat{\mathbf{H}}^{\mathrm{tru}}_u\hat{\mathbf{p}}_u = \mathbf{0}, \tag{21}$$

where $\mathbf{E}_{N_{\mathrm{e}}} = \left[\mathbf{I}_{\bar{N}_{\mathrm{e},u}}\right]_{N_{\mathrm{e}}:\bar{N}_{\mathrm{e},u}-1,:}$ is a selection matrix. Thus, $\bar{N}_{\mathrm{de},u}$ can be flexibly selected to adjust $\bar{N}_{\mathrm{e},u}$, and larger $\bar{N}_{\mathrm{de},u}$ can provide more degrees of freedom for optimization to achieve better performance at the cost of higher complexity.

According to (18), the space domain precoder of UT u at the k-th subcarrier is

$$\mathbf{p}_{u,k} = \left(\bar{\mathbf{u}}_{u,k}\otimes\mathbf{I}\right)\hat{\mathbf{p}}_u, \tag{22}$$

where $\bar{\mathbf{u}}_{u,k} = \left[\bar{\mathbf{U}}_u\right]_{k-k_0,:}$. Then the rate in (11) becomes

$$\mathcal{R}_{u,k} = \log\left(1+\frac{\mathbf{h}^{\mathrm{H}}_{u,k}\left(\bar{\mathbf{u}}_{u,k}\otimes\mathbf{I}\right)\hat{\mathbf{p}}_u\hat{\mathbf{p}}^{\mathrm{H}}_u\left(\bar{\mathbf{u}}^{\mathrm{H}}_{u,k}\otimes\mathbf{I}\right)\mathbf{h}_{u,k}}{\sigma^2_u+\sum\limits_{u'\in\mathcal{U}\setminus u}\mathbf{h}^{\mathrm{H}}_{u,k}\left(\bar{\mathbf{u}}_{u',k}\otimes\mathbf{I}\right)\hat{\mathbf{p}}_{u'}\hat{\mathbf{p}}^{\mathrm{H}}_{u'}\left(\bar{\mathbf{u}}^{\mathrm{H}}_{u',k}\otimes\mathbf{I}\right)\mathbf{h}_{u,k}}\right) \tag{23}$$

and the problem in (17) is transformed into

$$\begin{aligned}\left\{\hat{\mathbf{p}}^{\mathrm{op}}_u\right\}_{u\in\mathcal{U}} = &\arg\max_{\{\hat{\mathbf{p}}_u\}_{u\in\mathcal{U}}}\sum_{k\in\mathcal{K}}\sum_{u\in\mathcal{U}}w_u\mathcal{R}_{u,k}\\ &\text{s.t.}\quad\sum_{u\in\mathcal{U}}\hat{\mathbf{p}}^{\mathrm{H}}_u\left(\bar{\mathbf{U}}^{\mathrm{H}}_u\bar{\mathbf{U}}_u\otimes\mathbf{I}\right)\hat{\mathbf{p}}_u\le P,\\ &\quad\mathbf{E}_{N_{\mathrm{e}}}\hat{\mathbf{H}}^{\mathrm{tru}}_u\hat{\mathbf{p}}_u=\mathbf{0},\forall u\in\mathcal{U}.\end{aligned} \tag{24}$$

From (18) and (24), the design of the stacked precoder vector for a group of subcarriers, $\mathbf{p}_u$, is transformed to the design of $\hat{\mathbf{p}}_u$, whose dimension is $M\bar{N}_{\mathrm{de},u}$. Due to the large numbers of antennas and UTs in HF skywave massive MIMO-OFDM systems, the complexity of solving the problem in (24) may be unbearable in practical systems.

Fortunately, in HF skywave massive MIMO communications, the space beam domain channels typically exhibit sparsity due to the small angular spread [11], [38], [39], [40], therefore, most elements of the space beam domain channels are approximately zero. Such a channel sparsity can be exploited to reduce the complexity of the precoder design. The ordered index set of non-zero elements of the space beam domain channel for UT u is defined as $\mathcal{N}^{\mathrm{an}}_u = \left\{n_{\mathrm{an}}\middle|\left[\tilde{\mathbf{h}}^{\mathrm{H}}_{u,k}\right]_{n_{\mathrm{an}}}\neq 0\right\}$ with $N_{\mathrm{an},u} = |\mathcal{N}^{\mathrm{an}}_u|$, where the order is the index increasing order. Then we have the following theorem, proved in Appendix B.

Theorem 1: When $M\to\infty$ and $\forall u,u',\mathcal{N}^{\mathrm{an}}_u\cap\mathcal{N}^{\mathrm{an}}_{u'}=\emptyset$, the optimal solution of problem in (24) can be obtained as

$$\hat{\mathbf{p}}^{\mathrm{op}}_u = \left(\mathbf{I}_{\bar{N}_{\mathrm{de},u}}\otimes\mathbf{V}\mathbf{S}_u\right)\mathbf{q}^{\mathrm{op}}_u,\ \forall u\in\mathcal{U}, \tag{25}$$

where $\mathbf{S}_u\in\mathbb{R}^{N_{\mathrm{an}}\times N_{\mathrm{an},u}}$ is the beam selection matrix with

$$\left[\mathbf{S}_u\right]_{i,j} = \begin{cases}1, & i=\left[\mathcal{N}^{\mathrm{an}}_u\right]_j,\\ 0, & \text{otherwise},\end{cases} \tag{26}$$

and

$$\begin{aligned}\left\{\mathbf{q}^{\mathrm{op}}_u\right\}_{u\in\mathcal{U}} = &\arg\max_{\{\mathbf{q}_u\}_{u\in\mathcal{U}}}\sum_{k\in\mathcal{K}}\sum_{u\in\mathcal{U}}w_u\tilde{\mathcal{R}}_{u,k}\left(\mathbf{q}_u,\mathcal{N}^{\mathrm{an}}_u\right)\\ &\text{s.t.}\quad\sum_{u\in\mathcal{U}}\mathbf{q}^{\mathrm{H}}_u\left(\bar{\mathbf{U}}^{\mathrm{H}}_u\bar{\mathbf{U}}_u\otimes\mathbf{I}\right)\mathbf{q}_u\le P,\\ &\quad\mathbf{E}_{N_{\mathrm{e}}}\tilde{\mathbf{H}}^{\mathrm{tru}}_u\mathbf{q}_u=\mathbf{0},\forall u\in\mathcal{U},\end{aligned} \tag{27}$$

where

$$\tilde{\mathcal{R}}_{u,k}\left(\mathbf{q}_u,\mathcal{N}^{\mathrm{an}}_u\right) = \log\left(1+\frac{\bar{\mathbf{h}}^{\mathrm{H}}_{u,k}\left(\bar{\mathbf{u}}_{u,k}\otimes\mathbf{I}\right)\mathbf{q}_u\mathbf{q}^{\mathrm{H}}_u\left(\bar{\mathbf{u}}^{\mathrm{H}}_{u,k}\otimes\mathbf{I}\right)\bar{\mathbf{h}}_{u,k}}{\sigma^2_u+\sum\limits_{u'\in\mathcal{U}\setminus u}\bar{\mathbf{h}}^{\mathrm{H}}_{u,k}\left(\bar{\mathbf{u}}_{u,k}\otimes\mathbf{S}^{\mathrm{T}}_u\mathbf{S}_{u'}\right)\mathbf{q}_{u'}\mathbf{q}^{\mathrm{H}}_{u'}\left(\bar{\mathbf{u}}^{\mathrm{H}}_{u,k}\otimes\mathbf{S}^{\mathrm{T}}_{u'}\mathbf{S}_u\right)\bar{\mathbf{h}}_{u,k}}\right), \tag{28}$$

$$\tilde{\mathbf{H}}^{\mathrm{tru}}_u = \left[\frac{1}{\sqrt{N_{\mathrm{v}}}}\mathrm{blkcirc}\{\tilde{\mathbf{H}}^{\mathrm{H}}_u\mathbf{S}_u\}\right]_{0:\bar{N}_{\mathrm{e},u}-1,0:\bar{N}_{\mathrm{an},u}\bar{N}_{\mathrm{de},u}-1} \tag{29}$$

and $\bar{\mathbf{h}}^{\mathrm{H}}_{u,k} = \left[\tilde{\mathbf{h}}^{\mathrm{H}}_{u,k}\right]_{\mathcal{N}^{\mathrm{an}}_u}$.

From Theorem 1, for a sufficiently large M, the design of $\mathbf{p}_u\in\mathbb{C}^{MN_{\mathrm{v}}}$ is transformed into that of an SF beam domain vector $\mathbf{q}_u\in\mathbb{C}^{N_{\mathrm{an},u}N_{\mathrm{de},u}}$ in (27) when there are no overlapping space beam domain channels for any two UTs. Note that $N_{\mathrm{an},u}$ is the number of space beams and $N_{\mathrm{de},u}$ is the number of frequency beams. Moreover, due to the sparsity of the space beam domain channel in HF skywave massive MIMO communications, $N_{\mathrm{an},u}$ is typically much smaller than M, thus the complexity of the precoder design can be significantly reduced.

However, as the number of UTs increases, the condition that the space beam domain channels for any two UTs are non-overlapping may not be well satisfied. In this case, more space beams need to be considered to improve the sum-rate performance of the precoder. For this purpose, we have the following theorem, proved in Appendix C.

Theorem 2: Given space beam index sets $\mathcal{N}^{\mathrm{an}}_{u,1}\supseteq\mathcal{N}^{\mathrm{an}}_u,\forall u\in\mathcal{U}$, we add a new space beam for UT $\tilde{u}$, i.e., $\mathcal{N}^{\mathrm{an}}_{u,2}=\mathcal{N}^{\mathrm{an}}_{u,1},\forall u\in\mathcal{U}\setminus\tilde{u}$ and $\mathcal{N}^{\mathrm{an}}_{\tilde{u},2}=\mathcal{N}^{\mathrm{an}}_{\tilde{u},1}\cup\{n_{\mathrm{an},\tilde{u}}\}$. When $M\to\infty$, we have

$$\sum_{k\in\mathcal{K}}\sum_{u\in\mathcal{U}}w_u\tilde{\mathcal{R}}_{u,k}\left(\mathbf{q}^{\mathrm{op}}_u,\mathcal{N}^{\mathrm{an}}_{u,1}\right)\le\sum_{k\in\mathcal{K}}\sum_{u\in\mathcal{U}}w_u\tilde{\mathcal{R}}_{u,k}\left(\mathbf{q}^{\mathrm{op}}_u,\mathcal{N}^{\mathrm{an}}_{u,2}\right), \tag{30}$$

where equality holds provided that

$$n_{\mathrm{an},\tilde{u}}\notin\bigcup_{u:\mathcal{N}^{\mathrm{an}}_u\cap\mathcal{N}^{\mathrm{an}}_{\tilde{u},1}\neq\emptyset}\mathcal{N}^{\mathrm{an}}_u. \tag{31}$$

From Theorem 2, when the space beam domain channel of UT u overlaps with those of other UTs, it is not optimal to use only the space beams occupied by UT u, in the BSP design, and more beams need to be used for the precoder design. Theorem 2 provides a criterion for selecting the additional space beams for the precoder design. The space beams that satisfy condition (31) should not be selected since it will

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not improve the performance of the precoder while the space beams that satisfy the condition $n_{\mathrm{an},\tilde{u}} \in \bigcup_{u:\mathcal{N}_u^{\mathrm{an}} \cap \mathcal{N}_{\tilde{u},1}^{\mathrm{an}} \neq \emptyset} \mathcal{N}_u^{\mathrm{an}}$ can be selected to improve the performance. In other words, the added space beams are preferred to be selected from the UTs with overlapping space beam domain channel since it causes the inter-UT interference and should be considered when designing the precoder.

Although M is finite in practical, the above discussion suggests that we can design the precoder in the space beam domain for a group of subcarriers instead of in the space domain, which could reduces the complexity owing to the space beam domain channel sparsity in the HF skywave massive MIMO communications. Define a new space beam index set $\bar{\mathcal{N}}_u^{\mathrm{an}}$ with $\bar{N}_{\mathrm{an},u} = \left|\bar{\mathcal{N}}_u^{\mathrm{an}}\right|$. Motivated by (18) and (25), we constrain the stacked precoder vector for a group of subcarriers to have the beam structure as follows

$$\mathbf{p}_u = \left(\mathbf{U}_u \otimes \mathbf{V}_u\right) \mathbf{q}_u, \tag{32}$$

where $\mathbf{V}_u = \mathbf{V}\bar{\mathbf{S}}_u$ and $\bar{\mathbf{S}}_u \in \mathbb{R}^{N_{\mathrm{an}} \times \bar{N}_{\mathrm{an},u}}$ is the beam selection martrix with the space beam index set $\bar{\mathcal{N}}_u^{\mathrm{an}}$, i.e.,

$$\left[\bar{\mathbf{S}}_u\right]_{i,j} = \begin{cases} 1, & i = \left[\bar{\mathcal{N}}_u^{\mathrm{an}}\right]_j, \\ 0, & \text{otherwise.} \end{cases} \tag{33}$$

Then the rate in (11) becomes

$$\mathcal{R}_{u,k}\left(\mathbf{q}_u, \bar{\mathcal{N}}_u^{\mathrm{an}}\right) = \log\left(1 + \frac{\bar{\mathbf{h}}_{u,k}^{\mathrm{H}} \mathbf{Q}_{u,u,k} \mathbf{q}_u \mathbf{q}_u^{\mathrm{H}} \mathbf{Q}_{u,u,k}^{\mathrm{H}} \bar{\mathbf{h}}_{u,k}}{\sigma_u^2 + \sum\limits_{u' \in \mathcal{U} \setminus u} \bar{\mathbf{h}}_{u,k}^{\mathrm{H}} \mathbf{Q}_{u,u',k} \mathbf{q}_{u'} \mathbf{q}_{u'}^{\mathrm{H}} \mathbf{Q}_{u,u',k}^{\mathrm{H}} \bar{\mathbf{h}}_{u,k}}\right), \tag{34}$$

where $\mathbf{Q}_{u,u',k} = \mathbf{e}_k^{\mathrm{T}} \bar{\mathbf{U}}_{u'} \otimes \mathbf{S}_u^{\mathrm{T}} \mathbf{V}^{\mathrm{H}} \mathbf{V} \bar{\mathbf{S}}_{u'}$. We can turn to optimize $\mathbf{q}_u$ to maximize the sum-rate as follows

$$\begin{aligned} \left\{\mathbf{q}_u^{\mathrm{op}}\right\}_{u \in \mathcal{U}} = & \arg\max_{\{\mathbf{q}_u\}_{u \in \mathcal{U}}} \sum_{k \in \mathcal{K}} \sum_{u \in \mathcal{U}} w_u \mathcal{R}_{u,k}\left(\mathbf{q}_u, \bar{\mathcal{N}}_u^{\mathrm{an}}\right) \\ & \text{s.t.} \quad \sum_{u \in \mathcal{U}} \mathbf{q}_u^{\mathrm{H}} \mathbf{A}_u \mathbf{q}_u \leq P, \\ & \mathbf{E}_{N_{\mathrm{e}}} \tilde{\mathbf{H}}_u^{\mathrm{tru}} \left(\mathbf{I} \otimes \mathbf{S}_u^{\mathrm{T}} \mathbf{V}^{\mathrm{H}} \mathbf{V} \bar{\mathbf{S}}_{u'}\right) \mathbf{q}_u = \mathbf{0}, \forall u \in \mathcal{U}, \end{aligned} \tag{35}$$

where $\mathbf{A}_u = \bar{\mathbf{U}}_u^{\mathrm{H}} \bar{\mathbf{U}}_u \otimes \mathbf{V}_u^{\mathrm{H}} \mathbf{V}_u$. After obtaining $\mathbf{q}_u$ by solving the problem in (35), according to (32), the space domain precoder at each subcarrier is given by

$$\mathbf{p}_{u,k} = \mathbf{V}_u \left(\bar{\mathbf{u}}_{u,k} \otimes \mathbf{I}\right) \mathbf{q}_u = \mathbf{V}_u \tilde{\mathbf{q}}_{u,k}, \ \forall k \in \mathcal{K}. \tag{36}$$

where $\tilde{\mathbf{q}}_{u,k} \triangleq \left(\bar{\mathbf{u}}_{u,k} \otimes \mathbf{I}\right) \mathbf{q}_u$ is referred to as the space beam domain precoder at the k-th subcarrier, whose dimension is typically much smaller than that of $\mathbf{p}_{u,k}$ due to the space beam domain channel sparsity in HF skywave massive MIMO communications. Since only beam domain channel is involved in (35) and the space domain precoder $\mathbf{p}_{u,k}$ has the beam structure as shown in (36), i.e., the multiplication between the space beam matrix $\mathbf{V}_u$ and the space beam domain precoder $\tilde{\mathbf{q}}_{u,k}$, such a precoder design is referred to as the beam structured precoder (BSP).

From the channel smoothness constraint (i.e., the second constraint) in (35), $\mathbf{q}_u$ should lie in the null space of $\mathbf{E}_{N_{\mathrm{e}}} \tilde{\mathbf{H}}_u^{\mathrm{tru}} \left(\mathbf{I} \otimes \mathbf{S}_u^{\mathrm{T}} \mathbf{V}^{\mathrm{H}} \mathbf{V} \bar{\mathbf{S}}_{u'}\right)$, whose dimension is $\left(N_{\mathrm{de}} + \bar{N}_{\mathrm{de},u} - N_{\mathrm{e}} - 1\right) \times \bar{N}_{\mathrm{de},u} \bar{N}_{\mathrm{an},u}$. In this case, $\mathbf{q}_u$ can be rewritten as

$$\mathbf{q}_u = \bar{\mathbf{N}}_u \bar{\mathbf{q}}_u, \tag{37}$$

where $\bar{\mathbf{N}}_u \in \mathbb{C}^{\bar{N}_{\mathrm{de},u} \bar{N}_{\mathrm{an},u} \times N_{\mathrm{null},u}}$ is the matrix formed by bases of the null space of $\mathbf{E}_{N_{\mathrm{e}}} \tilde{\mathbf{H}}_u^{\mathrm{tru}} \left(\mathbf{I} \otimes \mathbf{S}_u^{\mathrm{T}} \mathbf{V}^{\mathrm{H}} \mathbf{V} \bar{\mathbf{S}}_{u'}\right)$ and can be obtained by SVD, and $N_{\mathrm{null},u} = \bar{N}_{\mathrm{de},u} \bar{N}_{\mathrm{an},u} - \left(N_{\mathrm{de}} + \bar{N}_{\mathrm{de},u} - N_{\mathrm{e}} - 1\right)$. By combining (32) and (37), $\mathbf{p}_u$ is further rewritten as

$$\mathbf{p}_u = \left(\mathbf{U}_u \otimes \mathbf{V}_u\right) \bar{\mathbf{N}}_u \bar{\mathbf{q}}_u, \tag{38}$$

which naturally satisfies the channel smoothness constraint in (35). Then the problem in (35) can be further transformed to the following problem

$$\begin{aligned} \left\{\bar{\mathbf{q}}_u^{\mathrm{op}}\right\}_{u \in \mathcal{U}} = & \arg\max_{\{\bar{\mathbf{q}}_u\}_{u \in \mathcal{U}}} \sum_{k \in \mathcal{K}} \sum_{u \in \mathcal{U}} w_u \mathcal{R}_{u,k}\left(\bar{\mathbf{q}}_u, \bar{\mathcal{N}}_u^{\mathrm{an}}\right) \\ & \text{s.t.} \quad \sum_{u \in \mathcal{U}} \bar{\mathbf{q}}_u^{\mathrm{H}} \bar{\mathbf{N}}_u^{\mathrm{H}} \mathbf{A}_u \bar{\mathbf{N}}_u \bar{\mathbf{q}}_u \leq P, \end{aligned} \tag{39}$$

where

$$\mathcal{R}_{u,k}\left(\bar{\mathbf{q}}_u, \bar{\mathcal{N}}_u^{\mathrm{an}}\right) = \log\left(1 + \frac{\bar{\mathbf{h}}_{u,k}^{\mathrm{H}} \mathbf{Q}_{u,u,k} \bar{\mathbf{N}}_u \bar{\mathbf{q}}_u \bar{\mathbf{q}}_u^{\mathrm{H}} \bar{\mathbf{N}}_u^{\mathrm{H}} \mathbf{Q}_{u,u,k}^{\mathrm{H}} \bar{\mathbf{h}}_{u,k}}{\sigma_u^2 + \sum\limits_{u' \in \mathcal{U} \setminus u} \bar{\mathbf{h}}_{u,k}^{\mathrm{H}} \mathbf{Q}_{u,u',k} \bar{\mathbf{N}}_{u'} \bar{\mathbf{q}}_{u'} \bar{\mathbf{q}}_{u'}^{\mathrm{H}} \bar{\mathbf{N}}_{u'}^{\mathrm{H}} \mathbf{Q}_{u,u',k}^{\mathrm{H}} \bar{\mathbf{h}}_{u,k}}\right). \tag{40}$$

In (39), the design of $\mathbf{q}_u$ in (35) with channel smoothness constraint is transformed to that of $\bar{\mathbf{q}}_u$ and only total power constraint is involved.

Next, we solve the problem in (39). According to [41, Lem. 1], we consider the following unconstrained problem and the relationship between the optimal solution $\bar{\mathbf{q}}_u^{\diamondsuit}$ of (41), as shown at the bottom of the next page and the optimal solution $\bar{\mathbf{q}}_u^{\mathrm{op}}$ of (39) is

$$\bar{\mathbf{q}}_u^{\mathrm{op}} = \sqrt{\frac{P}{\sum\limits_{u \in \mathcal{U}} \left(\bar{\mathbf{q}}_u^{\diamondsuit}\right)^{\mathrm{H}} \bar{\mathbf{N}}_u^{\mathrm{H}} \mathbf{A}_u \bar{\mathbf{N}}_u \bar{\mathbf{q}}_u^{\diamondsuit}}} \bar{\mathbf{q}}_u^{\diamondsuit}. \tag{42}$$

Therefore, we can solve the unconstrained optimization problem in (41) instead of the problem in (39), followed by a scaling operation. Moreover, the problem in (41) is equivalently transformed to the following WMMSE problem [24]

$$\max_{\{\alpha_{u,k}, d_{u,k}, \bar{\mathbf{q}}_u\}} \sum_{k \in \mathcal{K}} \sum_{u \in \mathcal{U}} w_u \left(\alpha_{u,k} \varepsilon_{u,k} - \log\left(\alpha_{u,k}\right)\right), \tag{43}$$

where

$$\begin{aligned} \varepsilon_{u,k} = & 1 - d_{u,k}^* \bar{\mathbf{h}}_{u,k}^{\mathrm{H}} \mathbf{Q}_{u,u,k} \bar{\mathbf{N}}_u \bar{\mathbf{q}}_u - \bar{\mathbf{q}}_u^{\mathrm{H}} \bar{\mathbf{N}}_u^{\mathrm{H}} \mathbf{Q}_{u,u,k}^{\mathrm{H}} \bar{\mathbf{h}}_{u,k} d_{u,k} \\ & + \sum_{u' \in \mathcal{U}} d_{u,k}^* d_{u,k} \bar{\mathbf{h}}_{u,k}^{\mathrm{H}} \mathbf{Q}_{u,u',k} \bar{\mathbf{N}}_{u'} \bar{\mathbf{q}}_{u'} \bar{\mathbf{q}}_{u'}^{\mathrm{H}} \bar{\mathbf{N}}_{u'}^{\mathrm{H}} \mathbf{Q}_{u,u',k}^{\mathrm{H}} \bar{\mathbf{h}}_{u,k} \\ & + d_{u,k}^* d_{u,k} \frac{\sigma_u^2}{P} \sum_{u' \in \mathcal{U}} \bar{\mathbf{q}}_{u'}^{\mathrm{H}} \bar{\mathbf{N}}_{u'}^{\mathrm{H}} \mathbf{A}_{u'} \bar{\mathbf{N}}_{u'} \bar{\mathbf{q}}_{u'}. \end{aligned} \tag{44}$$

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Then the block coordinate descent (BCD) is applied to solve the problem in (43). Specifically, at each iteration, we optimize one block of variables while fixing the others, and the subproblem with respect to each block variables can be solved by the first-order optimality condition while the convergence can be guaranteed. The resulting algorithm is summarized in Algorithm 1.

C. Efficient Calculation for Design and Implementation

The complexity of BSP per iteration in terms of the complex multiplication number is $\mathcal{O}\left(\sum_{u\in\mathcal{U}}\left(\bar{N}_{\mathrm{an},u}\bar{N}_{\mathrm{de},u}\right)^2 N_{\mathrm{null},u}+\sum_{u\in\mathcal{U}}\sum_{u'\in\mathcal{U}}\left(\bar{N}_{\mathrm{an},u'}+N_{\mathrm{null},u}\right)\bar{N}_{\mathrm{an},u}\bar{N}_{\mathrm{de},u}N_{\mathrm{v}}\right)$, which is still high in HF skywave massive MIMO-OFDM systems. In this subsection, we exploit the specific structure of the beam matrix and the beam structure of the precoder to propose the efficient calculation for design and implementation of BSP.

We first discuss an efficient design of BSP, i.e., an efficient calculation for Algorithm 1. Note that the elements of $\mathbf{V}^{\mathrm{H}}\mathbf{V}$ in $\mathbf{Q}_{u,u',k}$ can be given by

$$\left[\mathbf{V}^{\mathrm{H}}\mathbf{V}\right]_{n_{\mathrm{an}},n'_{\mathrm{an}}}=\Upsilon_M\left(\frac{2f_{\mathrm{c}}\Delta_\tau\left(n_{\mathrm{an}}-n'_{\mathrm{an}}\right)}{N_{\mathrm{an}}}\right), \tag{45}$$

where $\Upsilon_M(x)=e^{\bar{\jmath}\pi x\ (M-1)}\sin(\pi Mx)/(M\sin(\pi x))$. From (45), $\left|\left[\mathbf{V}^{\mathrm{H}}\mathbf{V}\right]_{n_{\mathrm{an}},n'_{\mathrm{an}}}\right|$ decreases rapidly as n'_{an} moves away from n_{an}, which implies the asymptotic orthogonality between the space beams. Thus for each UT, we define a set

$$\mathcal{T}_u=\left\{u'\left|\max_{i,j}\left|\left[\mathbf{S}_u^{\mathrm{T}}\mathbf{V}^{\mathrm{H}}\mathbf{V}\bar{\mathbf{S}}_{u'}\right]_{i,j}\right|>\eta\right.\right\}, \tag{46}$$

where η is a threshold. Then step 3 and step 5 in Algorithm 1 can be respectively approximated by

$$\begin{aligned}
&d_{u,k}\\
&\approx\left(\frac{\sigma_u^2}{P}\sum_{u'\in\mathcal{U}}\bar{\mathbf{q}}_{u'}^{\mathrm{H}}\bar{\mathbf{N}}_{u'}^{\mathrm{H}}\mathbf{A}_{u'}\bar{\mathbf{N}}_{u'}\bar{\mathbf{q}}_{u'}\right.\\
&\qquad\left.+\sum_{u'\in\mathcal{T}_u}\bar{\mathbf{h}}_{u,k}^{\mathrm{H}}\mathbf{Q}_{u,u',k}\bar{\mathbf{N}}_{u'}\bar{\mathbf{q}}_{u'}\bar{\mathbf{q}}_{u'}^{\mathrm{H}}\bar{\mathbf{N}}_{u'}^{\mathrm{H}}\mathbf{Q}_{u,u',k}^{\mathrm{H}}\bar{\mathbf{h}}_{u,k}\right)^{-1}\\
&\quad\times\bar{\mathbf{h}}_{u,k}^{\mathrm{H}}\mathbf{Q}_{u,u,k}\bar{\mathbf{N}}_u\bar{\mathbf{q}}_u,
\end{aligned} \tag{47}$$

$$\begin{aligned}
&\bar{\mathbf{q}}_u\\
&\approx\left(\sum_{u'\in\mathcal{T}_u}\sum_{k\in\mathcal{K}}w_{u'}\alpha_{u',k}d_{u',k}d_{u',k}^*\bar{\mathbf{N}}_u^{\mathrm{H}}\mathbf{Q}_{u',u,k}^{\mathrm{H}}\bar{\mathbf{h}}_{u',k}\bar{\mathbf{h}}_{u',k}^{\mathrm{H}}\mathbf{Q}_{u',u,k}\bar{\mathbf{N}}_u\right.\\
&\quad\left.+\sum_{u'\in\mathcal{U}}\sum_{k\in\mathcal{K}}\frac{\sigma_{u'}^2}{P}w_{u'}\alpha_{u',k}d_{u',k}d_{u',k}^*\bar{\mathbf{N}}_u^{\mathrm{H}}\mathbf{A}_u\bar{\mathbf{N}}_u\right)^{-1}\\
&\quad\times\sum_{k\in\mathcal{K}}w_u\alpha_{u,k}d_{u,k}\bar{\mathbf{N}}_u^{\mathrm{H}}\mathbf{Q}_{u,u,k}^{\mathrm{H}}\bar{\mathbf{h}}_{u,k}.
\end{aligned} \tag{48}$$

Note that $\mathcal{T}_u$ in (46) can be thought of as only depending on non-zero positions of the space beam domain channel of each UT, i.e., the statistical CSI of each UT, thus varies slowly and can be used for a relatively long period [11], [42].

Then we discuss the efficient calculation of (47). Specifically, multiplication $\mathbf{Q}_{u,u',k}\bar{\mathbf{N}}_{u'}\bar{\mathbf{q}}_{u'}$ is rewritten as

$$\mathbf{Q}_{u,u',k}\bar{\mathbf{N}}_{u'}\bar{\mathbf{q}}_{u'}=\left(\mathbf{e}_k^{\mathrm{T}}\otimes\mathbf{I}_{\bar{N}_{\mathrm{an},u}}\right)\left(\bar{\mathbf{U}}_{u'}\otimes\mathbf{S}_u^{\mathrm{T}}\mathbf{V}^{\mathrm{H}}\mathbf{V}\bar{\mathbf{S}}_{u'}\right)\bar{\mathbf{N}}_{u'}\bar{\mathbf{q}}_{u'}, \tag{49}$$

where $\bar{\mathbf{U}}_{u'}$ is a partial discret Fourier transform (DFT) matrix and the product of DFT matrices and the vector can be implemented efficiently by fast Fourier transform (FFT). Moreover, $\mathbf{S}_u^{\mathrm{T}}\mathbf{V}^{\mathrm{H}}\mathbf{V}\bar{\mathbf{S}}_{u'}$ and $\mathbf{A}_{u'}$ can be pre-calculated offline. Furthermore, term $\frac{\sigma_u^2}{P}\sum_{u'\in\mathcal{U}}\bar{\mathbf{q}}_{u'}^{\mathrm{H}}\bar{\mathbf{N}}_{u'}^{\mathrm{H}}\mathbf{A}_{u'}\bar{\mathbf{N}}_{u'}\bar{\mathbf{q}}_{u'}$ in (47) needs to be calculated only once for all UTs and a group of subcarriers. Thus the computational complexity of step 3 in terms of complex multiplication number is $\mathcal{O}\left(\sum_{u\in\mathcal{U}}\left(\bar{N}_{\mathrm{an},u}\bar{N}_{\mathrm{de},u}N_{\mathrm{null},u}+\bar{N}_{\mathrm{an},u}^2\bar{N}_{\mathrm{de},u}+\bar{N}_{\mathrm{an},u}\bar{N}_{\mathrm{de},u}^2\right)+\sum_{u\in\mathcal{U}}\sum_{u'\in\mathcal{T}_u}\left(N_{\mathrm{an},u}\bar{N}_{\mathrm{an},u'}\bar{N}_{\mathrm{de},u'}+N_{\mathrm{an},u}\bar{N}_{\mathrm{de}}\log_2\bar{N}_{\mathrm{de}}\right)\right)$.

Next, we discuss the efficient calculation of (48). First, the term $\sum_{u'\in\mathcal{U}}\sum_{k\in\mathcal{K}}\frac{\sigma_{u'}^2}{P}w_{u'}\alpha_{u',k}d_{u',k}d_{u',k}^*\bar{\mathbf{N}}_u^{\mathrm{H}}\mathbf{A}_u\bar{\mathbf{N}}_u$ needs to be calculated only once for all UTs and a group of subcarriers. Second, part of the calculation in (48) can be rewritten as

$$\begin{aligned}
&\sum_{k\in\mathcal{K}}w_{u'}\alpha_{u',k}d_{u',k}d_{u',k}^*\bar{\mathbf{N}}_u^{\mathrm{H}}\mathbf{Q}_{u',u,k}^{\mathrm{H}}\bar{\mathbf{h}}_{u',k}\bar{\mathbf{h}}_{u',k}^{\mathrm{H}}\mathbf{Q}_{u',u,k}\bar{\mathbf{N}}_u\\
&=\bar{\mathbf{N}}_u^{\mathrm{H}}\left(\sum_{k\in\mathcal{K}}w_{u'}\alpha_{u',k}d_{u',k}d_{u',k}^*\mathbf{Q}_{u',u,k}^{\mathrm{H}}\bar{\mathbf{h}}_{u',k}\bar{\mathbf{h}}_{u',k}^{\mathrm{H}}\mathbf{Q}_{u',u,k}\right)\bar{\mathbf{N}}_u.
\end{aligned} \tag{50}$$

Then we have

$$\begin{aligned}
&\sum_{k\in\mathcal{K}}w_{u'}\alpha_{u',k}d_{u',k}d_{u',k}^*\mathbf{Q}_{u',u,k}^{\mathrm{H}}\bar{\mathbf{h}}_{u',k}\bar{\mathbf{h}}_{u',k}^{\mathrm{H}}\mathbf{Q}_{u',u,k}\\
&=\sum_{k\in\mathcal{K}}\left(\bar{\mathbf{U}}_u^{\mathrm{H}}\mathbf{e}_k\otimes\mathbf{I}_{\bar{N}_{\mathrm{an},u}}\right)\mathbf{B}_{u,u',k}\left(\mathbf{e}_k^{\mathrm{T}}\bar{\mathbf{U}}_u\otimes\mathbf{I}_{\bar{N}_{\mathrm{an},u}}\right)\\
&=\left(\mathbf{I}_{\bar{N}_{\mathrm{de}}\times\bar{N}_{\mathrm{de},u}}^{\mathrm{T}}\otimes\mathbf{I}_{\bar{N}_{\mathrm{an},u}}\right)\left(\bar{\mathbf{U}}^{\mathrm{H}}\otimes\mathbf{I}_{\bar{N}_{\mathrm{an},u}}\right)\\
&\quad\times\mathrm{diag}\left\{\mathbf{B}_{u,u',k_0},\cdots,\mathbf{B}_{u,u',k_{N_{\mathrm{v}}-1}}\right\}\\
&\quad\times\left(\bar{\mathbf{U}}\otimes\mathbf{I}_{\bar{N}_{\mathrm{an},u}}\right)\left(\mathbf{I}_{\bar{N}_{\mathrm{de}}\times\bar{N}_{\mathrm{de},u}}\otimes\mathbf{I}_{\bar{N}_{\mathrm{an},u}}\right),
\end{aligned} \tag{51}$$

where $\mathbf{I}_{\bar{N}_{\mathrm{de}}\times\bar{N}_{\mathrm{de},u}}$ denotes the matrix composed of the first $\bar{N}_{\mathrm{de},u}$ columns of $\mathbf{I}_{\bar{N}_{\mathrm{de}}}$ and

$$\begin{aligned}
&\mathbf{B}_{u,u',k}\\
&=w_{u'}\alpha_{u',k}d_{u',k}d_{u',k}^*\bar{\mathbf{S}}_u^{\mathrm{T}}\mathbf{V}^{\mathrm{H}}\mathbf{V}\mathbf{S}_{u'}\bar{\mathbf{h}}_{u',k}\bar{\mathbf{h}}_{u',k}^{\mathrm{H}}\mathbf{S}_{u'}^{\mathrm{T}}\mathbf{V}^{\mathrm{H}}\mathbf{V}\bar{\mathbf{S}}_u.
\end{aligned} \tag{52}$$

Then we have

$$\left(\bar{\mathbf{U}}^{\mathrm{H}}\otimes\mathbf{I}_{\bar{N}_{\mathrm{an},u}}\right)\mathrm{diag}\left\{\mathbf{B}_{u,u',k_0},\cdots,\mathbf{B}_{u,u',k_{N_{\mathrm{v}}-1}}\right\}\left(\bar{\mathbf{U}}\otimes\mathbf{I}_{\bar{N}_{\mathrm{an},u}}\right)$$

$$\left\{\bar{\mathbf{q}}_u^{\diamondsuit}\right\}_{u\in\mathcal{U}}=\arg\max_{\{\bar{\mathbf{q}}_u\}_{u\in\mathcal{U}}}\sum_{k\in\mathcal{K}}\sum_{u\in\mathcal{U}}w_u\log\left(1+\frac{\bar{\mathbf{h}}_{u,k}^{\mathrm{H}}\mathbf{Q}_{u,u,k}\bar{\mathbf{N}}_u\bar{\mathbf{q}}_u\bar{\mathbf{q}}_u^{\mathrm{H}}\bar{\mathbf{N}}_u^{\mathrm{H}}\mathbf{Q}_{u,u,k}^{\mathrm{H}}\bar{\mathbf{h}}_{u,k}}{\frac{\sigma_u^2}{P}\sum_{u'\in\mathcal{U}}\bar{\mathbf{q}}_{u'}^{\mathrm{H}}\bar{\mathbf{N}}_{u'}^{\mathrm{H}}\mathbf{A}_{u'}\bar{\mathbf{N}}_{u'}\bar{\mathbf{q}}_{u'}+\sum_{u'\in\mathcal{U}\setminus u}\bar{\mathbf{h}}_{u,k}^{\mathrm{H}}\mathbf{Q}_{u,u',k}\bar{\mathbf{N}}_{u'}\bar{\mathbf{q}}_{u'}\bar{\mathbf{q}}_{u'}^{\mathrm{H}}\bar{\mathbf{N}}_{u'}^{\mathrm{H}}\mathbf{Q}_{u,u',k}^{\mathrm{H}}\bar{\mathbf{h}}_{u,k}}\right) \tag{41}$$

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Algorithm 1 BSP Design

Input: $\bar{\mathbf{h}}_{u,k}, \mathbf{Q}_{u,u',k}, \bar{\mathbf{N}}_u, \bar{\mathbf{S}}_u^{\mathrm{an}}, \sigma_u^2, \bar{N}_{\mathrm{de},u}, \forall u, u' \in \mathcal{U}, k \in \mathcal{K}$, N_{e}, P, iteration number T

Initialize: $\{\bar{\mathbf{q}}_u\}$ that satisfy $\sum_{u\in\mathcal{U}} \bar{\mathbf{q}}_u^{\mathrm{H}} \bar{\mathbf{N}}_u^{\mathrm{H}} \mathbf{A}_u \bar{\mathbf{N}}_u \bar{\mathbf{q}}_u = P$.

for $t = 1$ *to* T **do**

$\forall u, k$, update $d_{u,k}$:

$$d_{u,k} = \Big(\tfrac{\sigma_u^2}{P} \sum_{u'\in\mathcal{U}} \bar{\mathbf{q}}_{u'}^{\mathrm{H}} \bar{\mathbf{N}}_{u'}^{\mathrm{H}} \mathbf{A}_{u'} \bar{\mathbf{N}}_{u'} \bar{\mathbf{q}}_{u'} + \sum_{u'\in\mathcal{U}} \bar{\mathbf{h}}_{u,k}^{\mathrm{H}} \mathbf{Q}_{u,u',k} \bar{\mathbf{N}}_{u'} \bar{\mathbf{q}}_{u'} \bar{\mathbf{q}}_{u'}^{\mathrm{H}} \bar{\mathbf{N}}_{u'}^{\mathrm{H}} \mathbf{Q}_{u,u',k}^{\mathrm{H}} \bar{\mathbf{h}}_{u,k} \Big)^{-1} \bar{\mathbf{h}}_{u,k}^{\mathrm{H}} \mathbf{Q}_{u,u,k} \bar{\mathbf{N}}_u \bar{\mathbf{q}}_u$$

$\forall u, k$, update $\alpha_{u,k}$:

$$\alpha_{u,k} = \left(1 - d_{u,k}^* \bar{\mathbf{h}}_{u,k}^{\mathrm{H}} \mathbf{Q}_{u,u,k} \bar{\mathbf{N}}_u \bar{\mathbf{q}}_u\right)^{-1}$$

$\forall u$, update $\bar{\mathbf{q}}_u$:

$$\bar{\mathbf{q}}_u = \Big(\sum_{u'\in\mathcal{U}} \sum_{k\in\mathcal{K}} w_{u'} \alpha_{u',k} d_{u',k} d_{u',k}^* \bar{\mathbf{N}}_u^{\mathrm{H}} \mathbf{Q}_{u',u,k}^{\mathrm{H}} \bar{\mathbf{h}}_{u',k} \bar{\mathbf{h}}_{u',k}^{\mathrm{H}} \mathbf{Q}_{u',u,k} \bar{\mathbf{N}}_u + \sum_{u'\in\mathcal{U}} \sum_{k\in\mathcal{K}} \tfrac{\sigma_{u'}^2}{P} w_{u'} \alpha_{u',k} d_{u',k} d_{u',k}^* \bar{\mathbf{N}}_u^{\mathrm{H}} \mathbf{A}_u \bar{\mathbf{N}}_u \Big)^{-1} \sum_{k\in\mathcal{K}} w_u \alpha_{u,k} d_{u,k} \bar{\mathbf{N}}_u^{\mathrm{H}} \mathbf{Q}_{u,u,k}^{\mathrm{H}} \bar{\mathbf{h}}_{u,k}$$

end

Output: Scale $\{\bar{\mathbf{q}}_u\}$ to meet total power constraint. Then the space domain precoder at each subcarrier is obtained as

$$\mathbf{p}_{u,k} = \mathbf{V}_u \left(\bar{\mathbf{u}}_{u,k} \otimes \mathbf{I}\right) \bar{\mathbf{N}}_u \bar{\mathbf{q}}_u, \ \forall u \in \mathcal{U}, \forall k \in \mathcal{K}$$

$$= \mathrm{circ}\left\{\mathbf{C}_{u,u',0}, \cdots, \mathbf{C}_{u,u',\bar{N}_{\mathrm{de}}}\right\}, \tag{53}$$

where $\mathrm{circ}\left\{\mathbf{C}_{u,u',0}, \cdots, \mathbf{C}_{u,u',\bar{N}_{\mathrm{de}}}\right\}$ denotes the block-circulant matrix consists of $\mathbf{C}_{u,u',0}, \cdots, \mathbf{C}_{u,u',\bar{N}_{\mathrm{de}}}$, and

$$\left[\left[\mathbf{C}_{u,u',0}\right]_{a,b}, \cdots, \left[\mathbf{C}_{u,u',\bar{N}_{\mathrm{de}}}\right]_{a,b}\right]^{\mathrm{T}} = \frac{1}{\sqrt{\bar{N}_{\mathrm{de}}}} \bar{\mathbf{U}}^{\mathrm{H}} \left[\left[\mathbf{B}_{u,u',k_0}\right]_{a,b}, \cdots, \left[\mathbf{B}_{u,u',k_{N_{\mathrm{v}}-1}}\right]_{a,b}\right]^{\mathrm{T}}, \tag{54}$$

which can be calculated efficiently with FFT. Third, another part of the calculation in (48) can be rewritten as

$$\sum_{k\in\mathcal{K}} w_u \alpha_{u,k} d_{u,k} \bar{\mathbf{N}}_u^{\mathrm{H}} \mathbf{Q}_{u,u,k}^{\mathrm{H}} \bar{\mathbf{h}}_{u,k} = \bar{\mathbf{N}}_u^{\mathrm{H}} \sum_{k\in\mathcal{K}} w_u \alpha_{u,k} d_{u,k} \mathbf{Q}_{u,u,k}^{\mathrm{H}} \bar{\mathbf{h}}_{u,k} = \bar{\mathbf{N}}_u^{\mathrm{H}} \left(\bar{\mathbf{U}}_u^{\mathrm{H}} \otimes \mathbf{I}_{\bar{N}_{\mathrm{an},u}}\right) \left[\mathbf{b}_{u,k_0}^{\mathrm{T}}, \cdots, \mathbf{b}_{u,k_{N_{\mathrm{v}}-1}}^{\mathrm{T}}\right]^{\mathrm{T}}, \tag{55}$$

where $\mathbf{b}_{u,k} = w_u \alpha_{u,k} d_{u,k} \bar{\mathbf{S}}_u^{\mathrm{T}} \mathbf{V}^{\mathrm{H}} \mathbf{V} \mathbf{S}_u \bar{\mathbf{h}}_{u,k}$ and (55) can also be calculated efficiently with FFT. Therefore, the computational complexity of step 5 is $\mathcal{O}\left(\sum_{u\in\mathcal{U}} \sum_{u'\in\mathcal{T}_u} \left(\bar{N}_{\mathrm{an},u}^2 \bar{N}_{\mathrm{de}} \log_2 \bar{N}_{\mathrm{de}} + \left(\bar{N}_{\mathrm{an},u} \bar{N}_{\mathrm{de},u}\right)^2 N_{\mathrm{null},u}\right)\right)$.

In conclusion, since the complexity of Algorithm 1 is dominated by steps 3 and 5, with the efficient calculation discussed above, the total complexity per iteration is given by $\mathcal{O}\left(\sum_{u\in\mathcal{U}} \sum_{u'\in\mathcal{T}_u} \left(\bar{N}_{\mathrm{an},u} \bar{N}_{\mathrm{an},u'} \bar{N}_{\mathrm{de},u'} + \bar{N}_{\mathrm{an},u}^2 \bar{N}_{\mathrm{de}} \log_2 \bar{N}_{\mathrm{de}} + \left(\bar{N}_{\mathrm{an},u} \bar{N}_{\mathrm{an},u}\right)^2 N_{\mathrm{null},u}\right)\right)$.

After designing the precoder, we need to apply it to the data to generate the transmitted signal, which is referred to as the implementation of the BSP. Note that from (36), the transmitted signal at each subcarrier is given by

$$\sum_{u\in\mathcal{U}} \mathbf{p}_{u,k} x_{u,k} = \sum_{u\in\mathcal{U}} \mathbf{V}_u \tilde{\mathbf{q}}_{u,k} x_{u,k} = \mathbf{V} \sum_{u\in\mathcal{U}} \bar{\mathbf{S}}_u \tilde{\mathbf{q}}_{u,k} x_{u,k}. \tag{56}$$

Note that $\mathbf{V}$ can be regarded as a chirp z-transform (CZT) matrix, which can be rewritten as

$$\mathbf{V} = \mathrm{diag}\left\{\boldsymbol{\xi}_{W_{\mathrm{S}},N_{\mathrm{an}},0,M}\right\} \mathbf{F}_{N_{(\mathrm{S})}\times M}^{\mathrm{H}} \mathrm{diag}\left\{\tilde{\boldsymbol{\xi}}_{W_{\mathrm{S}},M,N_{\mathrm{an}},N_{(\mathrm{S})}}\right\} \mathbf{F}_{N_{(\mathrm{S})}\times N_{\mathrm{an}}} \mathrm{diag}\left\{\boldsymbol{\xi}_{W_{\mathrm{S}},0,0,N_{\mathrm{an}}}\right\}, \tag{57}$$

where $W_{\mathrm{S}} \triangleq e^{-\bar{\jmath}\frac{4\pi f_{\mathrm{c}} \Delta_\tau}{N_{\mathrm{an}}}}$. $\mathbf{F}_{N\times G}$ denotes the matrix composed of the first G ($\leq N$) columns of $\mathbf{F}_N$. $N_{(\mathrm{S})}$ is the integer greater than or equal to $M + N_{\mathrm{an}} - 1$. $\boldsymbol{\xi}_{W,\alpha_1,\alpha_2,N} \in \mathbb{C}^N$ with $\left[\boldsymbol{\xi}_{W,\alpha_1,\alpha_2,N}\right]_i = W^{(i^2-\alpha_1 i-\alpha_2)/2}$, and $\tilde{\boldsymbol{\xi}}_{W,N_1,N_2,N} \triangleq \sqrt{N}\mathbf{F}_N \left[\boldsymbol{\xi}_{W,0,0,N_1}^{\mathrm{H}}, \mathbf{0}, \boldsymbol{\xi}_{W,2(N_2-1),-(N_2-1)^2,N_2-1}^{\mathrm{H}}\right]^{\mathrm{T}} \in \mathbb{C}^N$. Thus the multiplication between matrix $\mathbf{V}$ and vector $\sum_{u\in\mathcal{U}} \bar{\mathbf{S}}_u \tilde{\mathbf{q}}_{u,k} x_{u,k}$ only consists of the product of diagonal matrices or DFT matrices and the vector, which can be implemented efficiently. Thus, the complexity of the efficient implementation is $\mathcal{O}\left(N_{(\mathrm{S})} \log_2 N_{(\mathrm{S})} + \sum_{u\in\mathcal{U}} \bar{N}_{\mathrm{an},u}\right)$.

IV. Effective Channel Estimation

Downlink effective channel estimation is critical for the subsequent data detection at each UT. In this section, we discuss the effective channel estimation at each UT when different precoders are applied at the BS. Pilot for downlink effective channel and the corresponding channel estimation approach for the proposed BSP are provided.

A. Pilots for Effective Channel

From (13), to perform the data detection at each UT in the DL transmission, the effective channel (i.e., the cascade of the channel and the precoder), instead of only the original channel, needs to be acquired. Thus for the effective channel estimation, the pilot should also be precoded with the same precoder for the data in the DL transmission.

We consider comb-type pilots and denote

$$\mathcal{K}_{\mathrm{p}} = \left\{k_0, k_0 + N_{\mathrm{comb}} - 1, k_0 + 2N_{\mathrm{comb}} - 1, \cdots, k_0 + (N_{\mathrm{p}} - 1) N_{\mathrm{comb}} - 1\right\} \subseteq \mathcal{K}$$

as the index set of subcarriers used for transmitting pilots. Denote $\mathbf{x}_u^{\mathrm{p}} \in \mathbb{C}^{N_{\mathrm{p}}}$ as the transmitted pilot signal for UT u and $\mathbf{X}_u^{\mathrm{p}} = \mathrm{diag}\left\{\mathbf{x}_u^{\mathrm{p}}\right\}$. Define $\mathbf{h}_{u,u}^{\mathrm{p}} = \left[\mathbf{h}_{u,u}^{\mathrm{e}}\right]_{\mathcal{K}_{\mathrm{p}}}$. Then with (13), the received pilot signal can be expressed as

$$\mathbf{y}_u^{\mathrm{p}} = \mathbf{X}_u^{\mathrm{p}} \mathbf{h}_{u,u}^{\mathrm{p}} + \sum_{u'\in\mathcal{U}\setminus u} \mathbf{X}_{u'}^{\mathrm{p}} \mathbf{h}_{u,u'}^{\mathrm{p}} + \mathbf{z}_u^{\mathrm{p}}. \tag{58}$$

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and $\mathbf{z}_u^{\mathrm{p}} = [\mathbf{z}_u]_{\mathcal{K}_{\mathrm{p}}}$. The widely used phase shift pilots [11], [43] can be selected as the pilot sequences, which is given by

$$\mathbf{x}_u^{\mathrm{p}} = \mathbf{x}_{\mathrm{c}} \circ \left[e^{-\bar{\jmath}2\pi k_0 \frac{\phi_u}{\bar{N}_{\mathrm{de}}}}, e^{-\bar{\jmath}2\pi k_{N_{\mathrm{comb}}-1} \frac{\phi_u}{\bar{N}_{\mathrm{de}}}}, \cdots, e^{-\bar{\jmath}2\pi k_{(N_{\mathrm{p}}-1)N_{\mathrm{comb}}-1} \frac{\phi_u}{\bar{N}_{\mathrm{de}}}}\right]^{\mathrm{T}}, \quad (59)$$

where $\phi_u \in \left\{0, N_{\mathrm{e}}, \cdots, \left\lfloor \bar{N}_{\mathrm{de}} / (N_{\mathrm{comb}} N_{\mathrm{e}}) \right\rfloor N_{\mathrm{e}}\right\}$ is referred to as the phase shift factor and $\mathbf{x}_{\mathrm{c}}$ is a sequence with unit modulus elements shared by all UTs.

B. Effective Channel Estimation

We first discuss the effective channel estimation when the precoder is designed at each subcarrier individually as shown in (12), where a well-known example is WMMSE precoder [24]. Since the channel smoothness constraint is not considered, one can only perform effective channel estimation at each pilot subcarrier independently and the least-square (LS) estimation can be used to obtain the estimation of the frequency domain effective channel of UT u, i.e.,

$$\hat{\mathbf{h}}_{u,u}^{\mathrm{p}} = (\mathbf{X}_u^{\mathrm{p}})^{-1} \mathbf{y}_u^{\mathrm{p}}. \quad (60)$$

The effective channel at data subcarriers can be obtained through interpolation with the channel estimates at the pilot subcarriers. Since the delay spread of the effective channel is not constrained, the effective channel is unsmooth and varies rapidly along the subcarriers, which implies that the obtained effective channel for the data subcarriers may be inaccurate.

We next discuss the effective channel estimation when the BSP is applied. Note that the channel smoothness constraint is considered in this case and the delay spread of the effective channel of UT u is constrained to N_{e}, that is, $\mathbf{h}_{u,u}^{\mathrm{p}}$ can be expressed as

$$\mathbf{h}_{u,u}^{\mathrm{p}} = \left[\bar{\mathbf{U}}\right]_{\mathcal{K}_{\mathrm{p}},0:N_{\mathrm{e}}-1} \bar{\mathbf{h}}_{u,u}^{\mathrm{p}}. \quad (61)$$

With (61) and the pilot in (59), the received pilot signal in (58) is rewritten as

$$\mathbf{y}_u^{\mathrm{p}} = \mathbf{X}_{\mathrm{c}} \bar{\mathbf{U}}_{\mathrm{p}} \mathbf{\Phi}_{u,u} \bar{\mathbf{h}}_{u,u}^{\mathrm{p}} + \sum_{u' \in \mathcal{U} \setminus u} \mathbf{X}_{u'}^{\mathrm{p}} \mathbf{h}_{u,u'}^{\mathrm{p}} + \mathbf{z}_u^{\mathrm{p}}, \quad (62)$$

where $\mathbf{X}_{\mathrm{c}} = \mathrm{diag}\{\mathbf{x}_{\mathrm{c}}\}$, $\bar{\mathbf{U}}_{\mathrm{p}} = \left[\bar{\mathbf{U}}\right]_{\mathcal{K}_{\mathrm{p}},:}$, $\mathbf{\Phi}_{u,u} = \left[\mathbf{I}_{\bar{N}_{\mathrm{de}}}\right]_{:,\phi_u N_{\mathrm{e}}:(\phi_u+1)N_{\mathrm{e}}-1}$. In this case, LS estimation can be applied to (62) to obtain the estimation of $\bar{\mathbf{h}}_{u,u}^{\mathrm{p}}$ as

$$\bar{\mathbf{h}}_{u,u}^{\mathrm{p,LS}} = \left(\mathbf{\Phi}_{u,u}^{\mathrm{T}} \bar{\mathbf{U}}_{\mathrm{p}}^{\mathrm{H}} \bar{\mathbf{U}}_{\mathrm{p}} \mathbf{\Phi}_{u,u}\right)^{-1} \mathbf{\Phi}_{u,u}^{\mathrm{T}} \bar{\mathbf{U}}_{\mathrm{p}}^{\mathrm{H}} \mathbf{X}_{\mathrm{c}}^{\mathrm{H}} \mathbf{y}_u^{\mathrm{p}}. \quad (63)$$

Then according to (61), the estimation of $\mathbf{h}_{u,u}^{\mathrm{p}}$ is given by

$$\mathbf{h}_{u,u}^{\mathrm{p,LS}} = \left[\bar{\mathbf{U}}\right]_{\mathcal{K}_{\mathrm{p}},0:N_{\mathrm{e}}-1} \bar{\mathbf{h}}_{u,u}^{\mathrm{p,LS}}, \quad (64)$$

and the estimation of $\mathbf{h}_{u,u}^{\mathrm{e}}$ including pilot and data subcarriers can be given by

$$\mathbf{h}_{u,u}^{\mathrm{e,LS}} = \left[\bar{\mathbf{U}}\right]_{:,0:N_{\mathrm{e}}-1} \bar{\mathbf{h}}_{u,u}^{\mathrm{p,LS}}. \quad (65)$$

In conclusion, the effective channel is smooth along the subcarriers since the delay spread of the effective channel is constrained, which facilitates the effective channel estimation at the data subcarriers.

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Carrier frequency f_{c}	16 MHz
Subcarrier spacing Δ_f	250 Hz
Number of subcarriers N_{c}	512
Length of CP N_{g}	128
Number of valid subcarriers N_{v}	384
Number of BS antennas M	256
BS antenna spacing d	9 m
Number of UTs U	64

V. SIMULATION RESULTS

In this section, we provide simulation results to illustrate the efficiency of the proposed BSP for HF skywave massive MIMO-OFDM communications.

A. Simulation Setup

The main simulation parameters are given in Table I. The BS is located at (34°N, 118°E). UTs are generated at a distance of 2,000 km from the BS and the azimuth angles of UTs seen from the BS are generated in the interval $(-90^\circ\ 90^\circ)$. We utilize a commercial ray-tracing software Proplab-Pro version 3.1 to generate the channel parameters of each UT independently [44]. Specifically, the software is used to obtain the parameters of each propagation path between the BS and each UT, including the path gain, the propagation delay, the azimuth and elevation AoAs. The initial phases and Doppler frequencies for paths are generated randomly. Then with the generated path parameters, the beam based channels can be generated according to (6) and (7), respectively, where fine factors, F_{an} and F_{de}, are set to two [11]. The noise power is set to be equal for all users. The priority weighted factors, $w_u, \forall u$ for all UTs are set to be 1. The power budgets for a group of subcarriers are set to be equal, i.e. $P_k = P/N_{\mathrm{v}}, \forall k \in \mathcal{K}$. The threshold for selecting interfering UTs, i.e., η in (46), is set to be 0.1.

Besides the proposed BSP, the WMMSE precoders designed at each subcarrier [24] are included for comparison. The initial values for the precoders are generated randomly. Moreover, the sum-rate in the simulartion is referred to the sum-rate for all UTs averaged over a group of subcarriers.

B. Sum-Rate Performance

Fig. 1 shows the sum-rate performance of BSP with different numbers of selected space beams, where $\bar{N}_{\mathrm{de},u} = 96$ and $N_{\mathrm{e}} = 64$. Note that to evaluate the sum-rate performance, we assume that perfect effective channel state information (ECSI) is available at each UT although this is not always guaranteed in practice. We select space beams for BSP according to Theorem 2. From the figure, the performance of the proposed BSP can approach that of WMMSE. Even though the performance gap increases with the transmit power due to the inter-UT interference, the gap can be reduced by including a higher number of space beams. This is because according to Theorem 2, we can use more space beams to improve the

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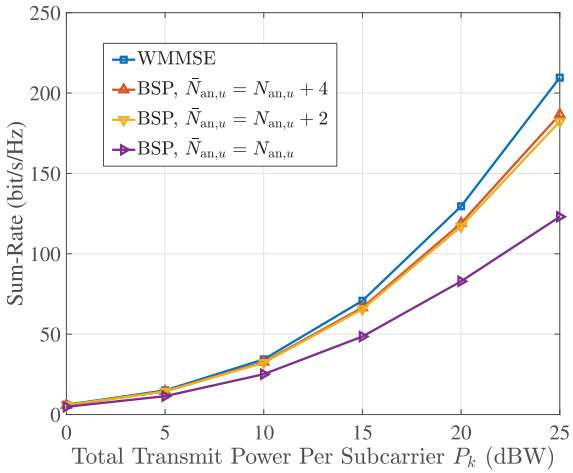


Fig. 1. Sum-rate performance of BSP with different numbers of selected space beams compared with WMMSE.

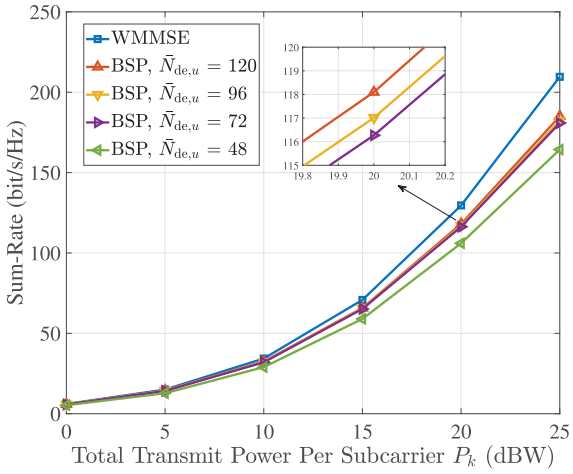


Fig. 2. Sum-rate performance of BSP with different numbers of used frequency beams compared with WMMSE.

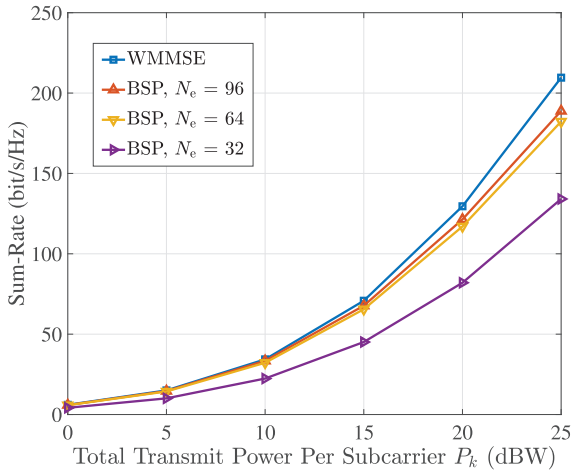


Fig. 3. Sum-rate performance of BSP with different constrained delay spread of the effective channel compared with WMMSE.

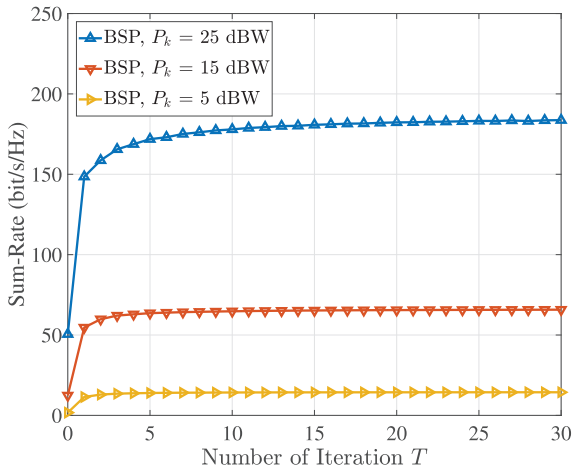


Fig. 4. Convergence of BSP with random initial values.

performance of BSP when there exists overlap between the space beam domain channels of different UTs.

Fig. 2 shows the sum-rate performance of BSP with different numbers of used frequency beams, where $\bar{N}_{\mathrm{an},u} = N_{\mathrm{an},u} + 2$ and $N_{\mathrm{e}} = 64$. From the figure, with the increase of the used frequency beams (i.e. $\bar{N}_{\mathrm{de},u}$), the performance gap can be reduced and the sum-rate performance of BSP can further approach that of WMMSE. This is because that larger $\bar{N}_{\mathrm{de},u}$ leads to a higher degree of freedom for optimization, which ultimately improves the sum-rate performance.

Fig. 3 shows the sum-rate performance of BSP with different constrained delay spread of the effective channel (i.e., N_{e}), where $\bar{N}_{\mathrm{an},u} = N_{\mathrm{an},u} + 2$ and $\bar{N}_{\mathrm{de},u} = 96$. From the figure, with the increase of N_{e}, the performance gap can be reduced and the sum-rate performance of BSP can further approach that of WMMSE. This is because that larger N_{e} results in a larger dimension of optimization variable $\bar{\mathbf{q}}_u$ as shown in (37), which provides more degrees of freedom for optimization and leads to better performance.

Fig. 4 shows the sum-rate performance of BSP at each iteration, where $\bar{N}_{\mathrm{an},u} = N_{\mathrm{an},u} + 2$, $\bar{N}_{\mathrm{de},u} = 96$ and $N_{\mathrm{e}} = 64$. From the figure, we observe that the proposed BSP exhibits fast convergence. Moreover, it is observed that as the transmit power increases, it takes more iterations to converge. This confirms the effectiveness of the proposed BSP for HF skywave massive MIMO-OFDM systems.

C. Effective Channel Estimation and BER Performance

In this subsection, we show the effective channel estimation performance with different precoders. The normalized mean-squared error (NMSE) is computed as

$$\mathrm{NMSE} = \frac{\sum\limits_{u\in\mathcal{U}} \left\| \mathbf{h}^{\mathrm{e}}_{u,u} - \mathbf{h}^{\mathrm{e,est}}_{u,u} \right\|^2}{\sum\limits_{u\in\mathcal{U}} \left\| \mathbf{h}^{\mathrm{e}}_{u,u} \right\|^2}, \tag{66}$$

where $\mathbf{h}^{\mathrm{e,est}}_{u,u}$ denotes the estimate of $\mathbf{h}^{\mathrm{e}}_{u,u}$ when any precoder is used.

Fig. 5 shows the NMSE performance of BSP with different constrained delay spreads of the effective channel, where

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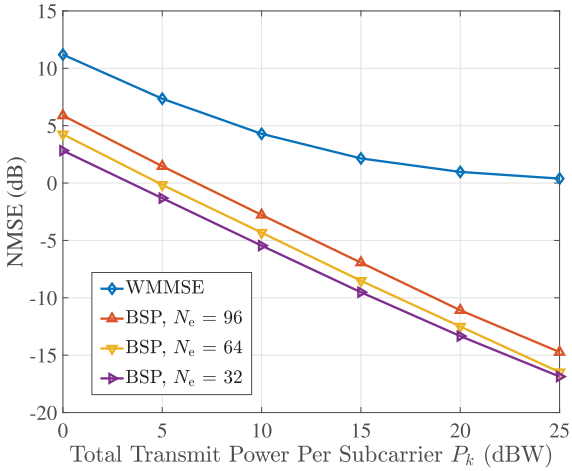


Fig. 5. NMSE performance of BSP with different constrained delay spread of the effective channel compared with WMMSE.

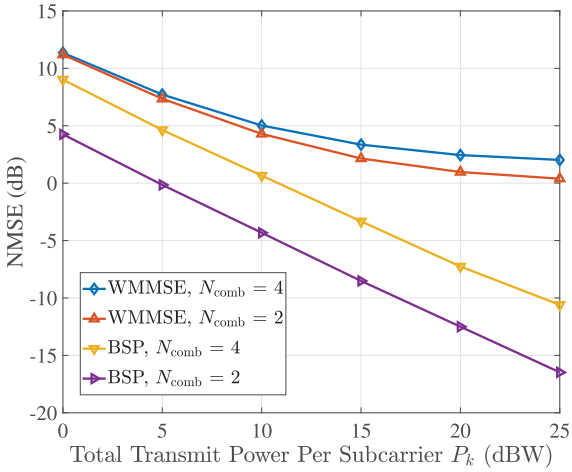


Fig. 6. NMSE performance of BSP and WMMSE with different N_{comb}.

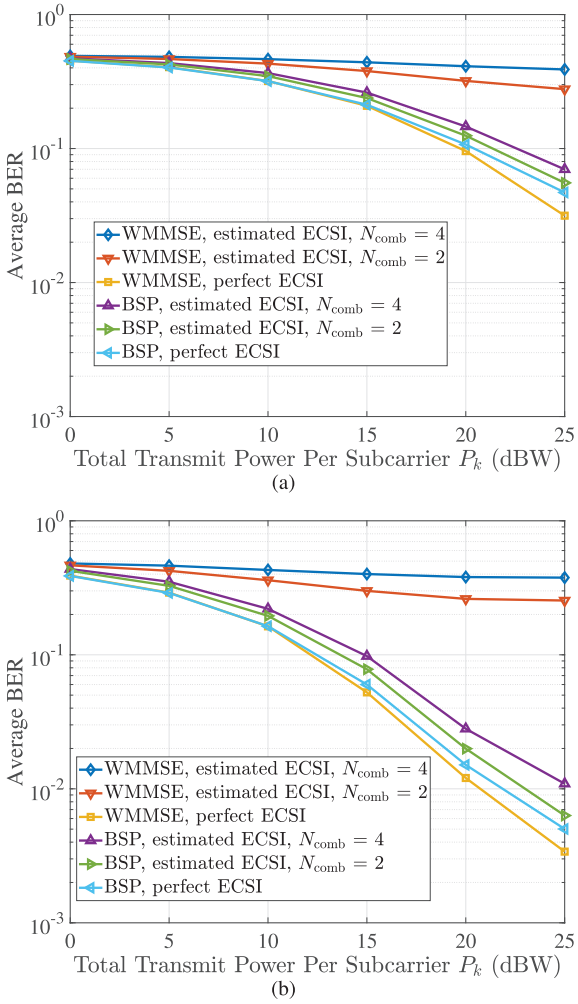


Fig. 7. BER performance of BSP and WMMSE with different N_{comb} using estimated effective channels. (a) $U = 64$; (b) $U = 32$.

$\bar{N}_{\text{an},u} = N_{\text{an},u} + 2$, $\bar{N}_{\text{de},u} = 96$ and $N_{\text{comb}} = 2$. From the figure, the effective channel estimation performance of BSP has a great advantage over that of WMMSE designed at each subcarrier, especially at the high transmit power region. This is because lower N_{e} means that the effective channel is smoother and varies more slowly along the subcarriers and the number of non-zero elements of the corresponding delay domain effective channel is fewer, which is conducive to the effective channel estimation. Furthermore, we observe that the performance gap between BSP with $N_{\text{e}} = 32$ and $N_{\text{e}} = 64$ becomes small as the transmit power increases. This is because that signal to interference plus noise ratio (SINR) with $N_{\text{e}} = 32$ is lower compared to that with $N_{\text{e}} = 64$, which can be observed from Fig. 3.

Fig. 6 shows the NMSE performance of BSP and WMMSE with different comb-type pilot configuration, where $\bar{N}_{\text{an},u} = N_{\text{an},u} + 2$, $\bar{N}_{\text{de},u} = 96$, $N_{\text{e}} = 64$. From the figure, with the increase of N_{comb}, the NMSE performance of BSP and WMMSE deteriorates since the number of pilots decreases. However, the NMSE performance of BSP can still outperform that of WMMSE.

Fig. 7 shows the BER performance of BSP and WMMSE with different comb-type pilot configurations using estimated effective channels, where $\bar{N}_{\text{an},u} = N_{\text{an},u} + 2$, $\bar{N}_{\text{de},u} = 96$, $N_{\text{e}} = 96$, the modulation scheme is quadrature phase shift keying (QPSK) and the zero-forcing detector is used. Average BER is referred to as the average of BER for all UTs. From the figure, with the estimated effective channel, BER performance of the proposed BSP has a great advantage over that of WMMSE thanks to the enhanced effective channel estimation performance. When the perfect ECSI is applied, BER performance of BSP can still approach that of WMMSE. Moreover, the performance gap between the WMMSE with estimated ECSI and perfect ECSI is large while the performance of BSP with estimated ECSI can approach that with perfect ECSI.

VI. Conclusion

In this paper, we have investigated precoder design for HF skywave massive MIMO-OFDM communications. We first revealed the effect of the precoder on the effective channel. Then we formulated the precoder design for a group of subcarriers as a sum-rate maximization problem, where

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the effective channel smoothness constraint was introduced. Moreover, the design of space domain precoders for a group of subcarriers was converted to that of an SF beam domain vector and the resulting space domain precoder at each subcarrier is beam structured. The corresponding space beam selection criterion was provided. We also presented the efficient calculation for design and implementation of the proposed BSP by exploiting the beam structure and the beam matrix. The downlink effective channel estimation with the proposed BSP was discussed. Simulation results demonstrated the advantages of the proposed BSP.

APPENDIX A
PROOF OF LEMMA 1

According to the beam based channel model in (7), the frequency beam domain (i.e. delay domain) channel at the m-th antenna is the m-th column of $\hat{\mathbf{H}}_u = \frac{1}{\sqrt{N_{\rm v}}}\tilde{\mathbf{H}}_u^{\rm H}\mathbf{V}^{\rm H} \in \mathbb{C}^{N_{\rm de}\times M}$, thus the DL space-frequency domain channel matrix $\bar{\mathbf{H}}_u^{\rm H}$ is rewritten as

$$\bar{\mathbf{H}}_u^{\rm H} = \bar{\mathbf{U}}\text{blkcirc}\{\hat{\mathbf{H}}_u\}\left(\bar{\mathbf{U}}^{\rm H}\otimes\mathbf{I}\right), \tag{67}$$

where $\text{blkcirc}\{\hat{\mathbf{H}}_u\} \in \mathbb{C}^{\bar{N}_{\rm de}\times M\bar{N}_{\rm de}}$. With (67) and (18), the effective channel of UT u can be rewritten as

$$\begin{aligned}\mathbf{h}_{u,u}^{\rm e} &= \bar{\mathbf{H}}_u^{\rm H}\mathbf{p}_u \\ &= \bar{\mathbf{U}}\text{blkcirc}\{\hat{\mathbf{H}}_u\}\left(\bar{\mathbf{U}}^{\rm H}\bar{\mathbf{U}}_u\otimes\mathbf{I}\right)\hat{\mathbf{p}}_u.\end{aligned} \tag{68}$$

By padding $\hat{\mathbf{p}}_u$ with zeros, we define that

$$\mathbf{p}_u^{\rm pad} = \left[\hat{\mathbf{p}}_u^{\rm T}, \mathbf{0}^{\rm T}\right]^{\rm T} \in \mathbb{C}^{M\bar{N}_{\rm de}}. \tag{69}$$

Since $\text{blkcirc}\{\hat{\mathbf{H}}_u\}$ is a block-circulant matrix, we have

$$\text{blkcirc}\{\hat{\mathbf{H}}_u\} = \mathbf{F}_{\bar{N}_{\rm de}}^{\rm H}\mathbf{\Lambda}\left(\mathbf{F}_{\bar{N}_{\rm de}}\otimes\mathbf{I}\right), \tag{70}$$

where $\mathbf{F}_N$ denotes the N-dimensional unitary DFT matrix, and $\mathbf{\Lambda}$ is a block-diagonal matrix. Then we have

$$\begin{aligned}&\bar{\mathbf{U}}\text{blkcirc}\{\hat{\mathbf{H}}_u\}\left(\bar{\mathbf{U}}^{\rm H}\bar{\mathbf{U}}_u\otimes\mathbf{I}\right)\hat{\mathbf{p}}_u \\ &= \bar{\mathbf{U}}\text{blkcirc}\{\hat{\mathbf{H}}_u\}\left(\bar{\mathbf{U}}^{\rm H}\bar{\mathbf{U}}\otimes\mathbf{I}\right)\mathbf{p}_u^{\rm pad} \\ &= \bar{\mathbf{U}}\mathbf{F}_{\bar{N}_{\rm de}}^{\rm H}\mathbf{\Lambda}\left(\mathbf{F}_{\bar{N}_{\rm de}}\otimes\mathbf{I}\right)\left(\bar{\mathbf{U}}^{\rm H}\bar{\mathbf{U}}\otimes\mathbf{I}\right)\left(\mathbf{F}_{\bar{N}_{\rm de}}^{\rm H}\otimes\mathbf{I}\right)\left(\mathbf{F}_{\bar{N}_{\rm de}}\otimes\mathbf{I}\right)\mathbf{p}_u^{\rm pad} \\ &= \bar{\mathbf{U}}\mathbf{F}_{\bar{N}_{\rm de}}^{\rm H}\mathbf{\Lambda}\left(\mathbf{F}_{\bar{N}_{\rm de}}\bar{\mathbf{U}}^{\rm H}\bar{\mathbf{U}}\mathbf{F}_{\bar{N}_{\rm de}}^{\rm H}\otimes\mathbf{I}\right)\left(\mathbf{F}_{\bar{N}_{\rm de}}\otimes\mathbf{I}\right)\mathbf{p}_u^{\rm pad} \\ &= \mathbf{I}_{N_{\rm v}\times\bar{N}_{\rm de}}\mathbf{\Lambda}\left(\mathbf{I}_{N_{\rm v}\times\bar{N}_{\rm de}}^{\rm H}\mathbf{I}_{N_{\rm v}\times\bar{N}_{\rm de}}\otimes\mathbf{I}\right)\left(\mathbf{F}_{\bar{N}_{\rm de}}\otimes\mathbf{I}\right)\mathbf{p}_u^{\rm pad} \\ &= \mathbf{I}_{N_{\rm v}\times\bar{N}_{\rm de}}\mathbf{\Lambda}\left(\mathbf{F}_{\bar{N}_{\rm de}}\otimes\mathbf{I}\right)\mathbf{p}_u^{\rm pad} \\ &= \bar{\mathbf{U}}\mathbf{F}_{\bar{N}_{\rm de}}^{\rm H}\mathbf{\Lambda}\left(\mathbf{F}_{\bar{N}_{\rm de}}\otimes\mathbf{I}\right)\mathbf{p}_u^{\rm pad} \\ &= \bar{\mathbf{U}}\text{blkcirc}\{\hat{\mathbf{H}}_u\}\mathbf{p}_u^{\rm pad},\end{aligned} \tag{71}$$

where $\mathbf{I}_{N_{\rm v}\times\bar{N}_{\rm de}}$ denotes the matrix composed of the first $N_{\rm v}$ rows of $\mathbf{I}_{\bar{N}_{\rm de}}$. With (15), (68) and (71), the delay domain effective channel can be given by

$$\tilde{\mathbf{h}}_{u,u}^{\rm e} = \text{blkcirc}\{\hat{\mathbf{H}}_u\}\mathbf{p}_u^{\rm pad}. \tag{72}$$

Moreover, the multiplication $\text{blkcirc}\{\hat{\mathbf{H}}_u\}\mathbf{p}_u^{\rm pad}$ is actually a vector-valued circular convolution between the first block column of $\text{blkcirc}\{\hat{\mathbf{H}}_u\}$ (i.e., $\left[\text{blkcirc}\{\hat{\mathbf{H}}_u\}\right]_{:,0:M-1}$) and $\mathbf{p}_u^{\rm pad}$, of which only the first $\min\left(N_{\rm de}+\bar{N}_{{\rm de},u}-1, \bar{N}_{\rm de}\right)$ elements can be non-zero. Therefore, we can obtain (19). This completes the proof.

APPENDIX B
PROOF OF THEOREM 1

Denote $\mathcal{N}^{\rm an} = \bigcup_{u\in\mathcal{U}}\mathcal{N}_u^{\rm an}$ with $\bar{N}_{\rm an} = |\mathcal{N}^{\rm an}|$, $\hat{\mathbf{V}} = [\mathbf{V}]_{:,\mathcal{N}^{\rm an}}$ and $\hat{\mathbf{h}}_{u,k} = \left[\tilde{\mathbf{h}}_{u,k}\right]_{\mathcal{N}^{\rm an}}$. Define $\hat{\mathbf{q}}_u = \left(\mathbf{I}\otimes\hat{\mathbf{V}}^{\rm H}\right)\hat{\mathbf{p}}_u \in \mathbb{C}^{\bar{N}_{\rm an}\bar{N}_{{\rm de},u}}$, then with the beam based channel model in (6), $\mathcal{R}_{u,k}$ in (23) is rewritten as

$$\begin{aligned}&\mathcal{R}_{u,k} \\ &= \log\left(1+\frac{\hat{\mathbf{h}}_{u,k}^{\rm H}\left(\bar{\mathbf{u}}_{u,k}\otimes\mathbf{I}\right)\hat{\mathbf{q}}_u\hat{\mathbf{q}}_u^{\rm H}\left(\bar{\mathbf{u}}_{u,k}^{\rm H}\otimes\mathbf{I}\right)\hat{\mathbf{h}}_{u,k}}{\sigma_u^2+\sum\limits_{u'\in\mathcal{U}\setminus u}\hat{\mathbf{h}}_{u,k}^{\rm H}\left(\bar{\mathbf{u}}_{u',k}\otimes\mathbf{I}\right)\hat{\mathbf{q}}_{u'}\hat{\mathbf{q}}_{u'}^{\rm H}\left(\bar{\mathbf{u}}_{u',k}^{\rm H}\otimes\mathbf{I}\right)\hat{\mathbf{h}}_{u,k}}\right).\end{aligned} \tag{73}$$

Note that to ensure the existence of $\hat{\mathbf{p}}_u$ such that $\hat{\mathbf{q}}_u = \left(\mathbf{I}\otimes\hat{\mathbf{V}}^{\rm H}\right)\hat{\mathbf{p}}_u$ when $\hat{\mathbf{q}}_u$ is given, $\hat{\mathbf{q}}_u$ should satisfy the constraint $\left(\mathbf{I}\otimes\hat{\mathbf{V}}^{\rm H}\left(\hat{\mathbf{V}}^{\rm H}\right)^{\dagger}\right)\hat{\mathbf{q}}_u = \hat{\mathbf{q}}_u$ [45]. Since $\lim\limits_{M\to\infty}\hat{\mathbf{V}}^{\rm H}\hat{\mathbf{V}} = \mathbf{I}$, when $M\to\infty$, such a constraint is always satisfied. Thus we can turn to optimize $\hat{\mathbf{q}}_u$ to maximize the sum-rate. In this case, the channel smoothness constraint is rewritten as

$$\mathbf{E}_{N_{\rm e}}\tilde{\mathbf{H}}_u^{\rm tru}\left(\mathbf{I}\otimes\mathbf{I}_{\bar{N}_{{\rm an},u}\times\bar{N}_{\rm an}}\right)\hat{\mathbf{q}}_u = \mathbf{0}. \tag{74}$$

We next discuss how to generate the power constraint with respect to $\hat{\mathbf{q}}_u$. Note that for the optimal $\hat{\mathbf{q}}_u^{\rm op}$, the corresponding optimal $\mathbf{p}_u^{\rm op}$ should have the minimal power, i.e., $\left(\mathbf{p}_u^{\rm op}\right)^{\rm H}\mathbf{p}_u^{\rm op}$ should be minimal. And the total power, $\sum\limits_{u\in\mathcal{U}}\left(\mathbf{p}_u^{\rm op}\right)^{\rm H}\mathbf{p}_u^{\rm op}$, should be equal to P in this case. Otherwise, $\mathbf{p}_u^{\rm op}, u\in\mathcal{U}$, can be scaled up to increase the sum-rate and thus it contradicts the optimality. Thus, we can formulate the following optimization problem

$$\begin{aligned}\mathbf{p}_u^{\rm op} = \ &\underset{\mathbf{p}_u}{\text{argmin}}\ \mathbf{p}_u^{\rm H}\mathbf{p}_u \\ &\text{s.t. }\left(\mathbf{I}\otimes\hat{\mathbf{V}}^{\rm H}\right)\mathbf{p}_u = \left(\bar{\mathbf{U}}_u\otimes\mathbf{I}\right)\hat{\mathbf{q}}_u^{\rm op}.\end{aligned} \tag{75}$$

The optimal solution of (75) is obtained as

$$\mathbf{p}_u^{\rm op} = \left(\bar{\mathbf{U}}_u\otimes\hat{\mathbf{V}}\left(\hat{\mathbf{V}}^{\rm H}\hat{\mathbf{V}}\right)^{\dagger}\right)\hat{\mathbf{q}}_u^{\rm op} = \left(\bar{\mathbf{U}}_u\otimes\hat{\mathbf{V}}\right)\hat{\mathbf{q}}_u^{\rm op}. \tag{76}$$

By combining (18) and (76), we can obtain that

$$\hat{\mathbf{p}}_u^{\rm op} = \left(\mathbf{I}\otimes\hat{\mathbf{V}}\right)\hat{\mathbf{q}}_u^{\rm op}. \tag{77}$$

Then by replacing $\hat{\mathbf{p}}_u$ with $\left(\mathbf{I}\otimes\hat{\mathbf{V}}\right)\hat{\mathbf{q}}_u^{\rm op}$, the total power constraint in problem (24) becomes

$$\sum_{u\in\mathcal{U}}\hat{\mathbf{q}}_u^{\rm H}\left(\bar{\mathbf{U}}_u^{\rm H}\bar{\mathbf{U}}_u\otimes\mathbf{I}\right)\hat{\mathbf{q}}_u \le P. \tag{78}$$

Next, we prove that elements of $\hat{\mathbf{q}}_u^{\rm op}$ are non-zero only at the positions corresponding to $\mathcal{N}_u^{\rm an}$. First,we define $\mathbf{q}_u^{\rm op} = \hat{\mathbf{q}}_u^{\rm op}\circ\left(\mathbf{1}_{\bar{N}_{{\rm de},\tilde{u}}}\otimes\hat{\mathbf{n}}_u\right)$, where

$$[\hat{\mathbf{n}}_u]_i = \begin{cases}1, & [\mathcal{N}^{\rm an}]_i \in \mathcal{N}_u^{\rm an},\\ 0, & \text{otherwise}.\end{cases} \tag{79}$$

It can be checked that $\left(\mathbf{q}_u^{\rm op}\right)^{\rm H}\left(\bar{\mathbf{U}}_u^{\rm H}\bar{\mathbf{U}}_u\otimes\mathbf{I}\right)\mathbf{q}_u^{\rm op} \le \left(\hat{\mathbf{q}}_u^{\rm op}\right)^{\rm H}\left(\bar{\mathbf{U}}_u^{\rm H}\bar{\mathbf{U}}_u\otimes\mathbf{I}\right)\hat{\mathbf{q}}_u^{\rm op}$, thus $\mathbf{q}_u^{\rm op}$ satisfies the total power constraint in (78). Moreover, we have

$$\begin{aligned}&\mathbf{E}_{N_{\rm e}}\tilde{\mathbf{H}}_u^{\rm tru}\left(\mathbf{I}\otimes\mathbf{I}_{\bar{N}_{{\rm an},u}\times\bar{N}_{\rm an}}\right)\mathbf{q}_u^{\rm op} \\ &= \mathbf{E}_{N_{\rm e}}\tilde{\mathbf{H}}_u^{\rm tru}\left(\mathbf{I}\otimes\mathbf{I}_{\bar{N}_{{\rm an},u}\times\bar{N}_{\rm an}}\right)\hat{\mathbf{q}}_u^{\rm op} = \mathbf{0},\end{aligned} \tag{80}$$

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which means that $\mathbf{q}_u^{\mathrm{op}}$ also satisfies channel smoothness constraint in (74). Furthermore, since $\hat{\mathbf{q}}_u^{\mathrm{op}}$ is the optimal solution, we have $\sum_{k\in\mathcal{K}}\sum_{u\in\mathcal{U}} w_u\mathcal{R}_{u,k}\left(\mathbf{q}_u^{\mathrm{op}}\right) \le \sum_{k\in\mathcal{K}}\sum_{u\in\mathcal{U}} w_u\mathcal{R}_{u,k}\left(\hat{\mathbf{q}}_u^{\mathrm{op}}\right)$. On the other hand, when $\forall u,u',\mathcal{N}_u^{\mathrm{an}}\cap\mathcal{N}_{u'}^{\mathrm{an}}=\emptyset$ and $\hat{\mathbf{q}}_u^{\mathrm{op}}$ is replaced by $\mathbf{q}_u^{\mathrm{op}}$ when calculating the sum-rate using (73), we can find that the numerator stays the same and the denominator may be the same or decrease in (73), which leads to $\sum_{k\in\mathcal{K}}\sum_{u\in\mathcal{U}} w_u\mathcal{R}_{u,k}\left(\mathbf{q}_u^{\mathrm{op}}\right) \ge \sum_{k\in\mathcal{K}}\sum_{u\in\mathcal{U}} w_u\mathcal{R}_{u,k}\left(\hat{\mathbf{q}}_u^{\mathrm{op}}\right)$. Therefore we can conclude that

$$\sum_{k\in\mathcal{K}}\sum_{u\in\mathcal{U}} w_u\mathcal{R}_{u,k}\left(\mathbf{q}_u^{\mathrm{op}}\right) = \sum_{k\in\mathcal{K}}\sum_{u\in\mathcal{U}} w_u\mathcal{R}_{u,k}\left(\hat{\mathbf{q}}_u^{\mathrm{op}}\right), \tag{81}$$

which indicates that we can focus on the non-zero elements of $\hat{\mathbf{q}}_u^{\mathrm{op}}$. In this case, (77) becomes (25), $\mathcal{R}_{u,k}$ in (73) becomes that in (28), and constrains (78) and (74) become constraints in (27). This completes the proof.

Appendix C
Proof of Theorem 2

Since the case of smaller index sets, $\mathcal{N}_{u,1}^{\mathrm{an}},\forall u\in\mathcal{U}$, is a special case of larger index sets, $\mathcal{N}_{u,2}^{\mathrm{an}},\forall u\in\mathcal{U}$, we can directly obtain that

$$\sum_{k\in\mathcal{K}}\sum_{u\in\mathcal{U}} w_u\tilde{\mathcal{R}}_{u,k}\left(\mathbf{q}_u^{\mathrm{op}},\mathcal{N}_{u,1}^{\mathrm{an}}\right) \le \sum_{k\in\mathcal{K}}\sum_{u\in\mathcal{U}} w_u\tilde{\mathcal{R}}_{u,k}\left(\mathbf{q}_u^{\mathrm{op}},\mathcal{N}_{u,2}^{\mathrm{an}}\right). \tag{82}$$

Denote the optimal solution with the space beam index sets $\mathcal{N}_{u,2}^{\mathrm{an}},\forall u\in\mathcal{U}$ as $\mathbf{q}_u^{\diamondsuit},\forall u\in\mathcal{U}$. Then for UT $\tilde{u}$, we denote $\mathbf{q}_{\tilde{u}}^{\circ}=\mathbf{q}_{\tilde{u}}^{\diamondsuit}\circ\left(\mathbf{1}_{\bar{N}_{\mathrm{de},\tilde{u}}}\otimes\mathbf{n}_{\tilde{u}}\right)$, where

$$\left[\mathbf{n}_{\tilde{u}}\right]_i=\begin{cases}1, & \left[\mathcal{N}_{\tilde{u},2}^{\mathrm{an}}\right]_i\in\mathcal{N}_{\tilde{u},1}^{\mathrm{an}},\\ 0, & \text{otherwise},\end{cases} \tag{83}$$

and $\mathbf{q}_u^{\circ}=\mathbf{q}_u^{\diamondsuit},\forall u\in\mathcal{U}\backslash\tilde{u}$. It can be checked that $\left(\mathbf{q}_u^{\circ}\right)^{\mathrm{H}}\left(\bar{\mathbf{U}}_u^{\mathrm{H}}\bar{\mathbf{U}}_u\otimes\mathbf{I}\right)\mathbf{q}_u^{\circ} \le \left(\mathbf{q}_u^{\diamondsuit}\right)^{\mathrm{H}}\left(\bar{\mathbf{U}}_u^{\mathrm{H}}\bar{\mathbf{U}}_u\otimes\mathbf{I}\right)\mathbf{q}_u^{\diamondsuit}$, thus $\mathbf{q}_u^{\circ},\forall u\in\mathcal{U}$ satisfies the total power constraint. Moreover, we have

$$\begin{aligned}&\mathbf{E}_{N_{\mathrm{e}}}\tilde{\mathbf{H}}_u^{\mathrm{tru}}\left(\mathbf{I}\otimes\left[\mathbf{V}\right]_{:,\mathcal{N}_u^{\mathrm{an}}}^{\mathrm{H}}\left[\mathbf{V}\right]_{:,\mathcal{N}_{u,2}^{\mathrm{an}}}\right)\mathbf{q}_u^{\circ}\\ &=\mathbf{E}_{N_{\mathrm{e}}}\tilde{\mathbf{H}}_u^{\mathrm{tru}}\left(\mathbf{I}\otimes\mathbf{I}_{\bar{N}_{\mathrm{an},u}\times\left|\mathcal{N}_{u,2}^{\mathrm{an}}\right|}\right)\mathbf{q}_u^{\diamondsuit}=\mathbf{0},\end{aligned} \tag{84}$$

which means that $\mathbf{q}_u^{\circ},\forall u\in\mathcal{U}$, also satisfies channel smoothness constraint. Define $\bar{\mathbf{Q}}_{u,u',k}=\mathbf{e}_k^{\mathrm{T}}\bar{\mathbf{U}}_{u'}\otimes\left[\mathbf{I}_{N_{\mathrm{an}}}\right]_{\mathcal{N}_u^{\mathrm{an}},\mathcal{N}_{u',2}^{\mathrm{an}}}$, then we have

$$\begin{aligned}&\left[\tilde{\mathbf{h}}_{u,k}^{\mathrm{H}}\right]_{\mathcal{N}_u^{\mathrm{an}}}\bar{\mathbf{Q}}_{u,u,k}\mathbf{q}_u^{\circ}\left(\mathbf{q}_u^{\circ}\right)^{\mathrm{H}}\bar{\mathbf{Q}}_{u,u,k}^{\mathrm{H}}\left[\tilde{\mathbf{h}}_{u,k}\right]_{\mathcal{N}_u^{\mathrm{an}}}\\ &=\left[\tilde{\mathbf{h}}_u^{\mathrm{H}}\right]_{\mathcal{N}_{u,2}^{\mathrm{an}}}\bar{\mathbf{Q}}_{u,u,k}\mathbf{q}_u^{\diamondsuit}\left(\mathbf{q}_u^{\diamondsuit}\right)^{\mathrm{H}}\bar{\mathbf{Q}}_{u,u,k}^{\mathrm{H}}\left[\tilde{\mathbf{h}}_{u,k}\right]_{\mathcal{N}_u^{\mathrm{an}}},\end{aligned} \tag{85}$$

which leads to

$$\tilde{\mathcal{R}}_{\tilde{u},k}\left(\mathbf{q}_{\tilde{u}}^{\circ},\mathcal{N}_{\tilde{u},2}^{\mathrm{an}}\right)=\tilde{\mathcal{R}}_{\tilde{u},k}\left(\mathbf{q}_{\tilde{u}}^{\diamondsuit},\mathcal{N}_{\tilde{u},2}^{\mathrm{an}}\right). \tag{86}$$

Moreover, for $u\neq\tilde{u}$, when $n_{\mathrm{an},\tilde{u}}\notin\bigcup_{u:\mathcal{N}_u^{\mathrm{an}}\cap\mathcal{N}_{\tilde{u},1}^{\mathrm{an}}\neq\emptyset}\mathcal{N}_u^{\mathrm{an}}$, we have

$$\begin{aligned}&\left[\tilde{\mathbf{h}}_{u,k}^{\mathrm{H}}\right]_{\mathcal{N}_{u,2}^{\mathrm{an}}}\bar{\mathbf{Q}}_{u,u',k}\mathbf{q}_{u'}^{\circ}\left(\mathbf{q}_{u'}^{\circ}\right)^{\mathrm{H}}\bar{\mathbf{Q}}_{u,u',k}^{\mathrm{H}}\left[\tilde{\mathbf{h}}_{u,k}\right]_{\mathcal{N}_{u,2}^{\mathrm{an}}}\\ &\le\left[\tilde{\mathbf{h}}_{u,k}^{\mathrm{H}}\right]_{\mathcal{N}_{u,2}^{\mathrm{an}}}\bar{\mathbf{Q}}_{u,u',k}\mathbf{q}_{u'}^{\diamondsuit}\left(\mathbf{q}_{u'}^{\diamondsuit}\right)^{\mathrm{H}}\bar{\mathbf{Q}}_{u,u',k}^{\mathrm{H}}\left[\tilde{\mathbf{h}}_{u,k}\right]_{\mathcal{N}_{u,2}^{\mathrm{an}}},\end{aligned} \tag{87}$$

which leads to

$$\sum_{k\in\mathcal{K}}\sum_{u\in\mathcal{U}\backslash\tilde{u}} w_u\tilde{\mathcal{R}}_{\tilde{u},k}\left(\mathbf{q}_{\tilde{u}}^{\circ},\mathcal{N}_{\tilde{u},2}^{\mathrm{an}}\right) \ge \sum_{k\in\mathcal{K}}\sum_{u\in\mathcal{U}\backslash\tilde{u}} w_u\tilde{\mathcal{R}}_{\tilde{u},k}\left(\mathbf{q}_{\tilde{u}}^{\diamondsuit},\mathcal{N}_{\tilde{u},2}^{\mathrm{an}}\right). \tag{88}$$

By combining (86) and (88), we have

$$\sum_{k\in\mathcal{K}}\sum_{u\in\mathcal{U}} w_u\tilde{\mathcal{R}}_{\tilde{u},k}\left(\mathbf{q}_{\tilde{u}}^{\circ},\mathcal{N}_{\tilde{u},2}^{\mathrm{an}}\right) \ge \sum_{k\in\mathcal{K}}\sum_{u\in\mathcal{U}} w_u\tilde{\mathcal{R}}_{\tilde{u},k}\left(\mathbf{q}_{\tilde{u}}^{\diamondsuit},\mathcal{N}_{\tilde{u},2}^{\mathrm{an}}\right). \tag{89}$$

By combining (82) and (89), we have

$$\sum_{k\in\mathcal{K}}\sum_{u\in\mathcal{U}} w_u\tilde{\mathcal{R}}_{\tilde{u},k}\left(\mathbf{q}_{\tilde{u}}^{\circ},\mathcal{N}_{\tilde{u},2}^{\mathrm{an}}\right) = \sum_{k\in\mathcal{K}}\sum_{u\in\mathcal{U}} w_u\tilde{\mathcal{R}}_{\tilde{u},k}\left(\mathbf{q}_{\tilde{u}}^{\diamondsuit},\mathcal{N}_{\tilde{u},2}^{\mathrm{an}}\right). \tag{90}$$

which indicates that $\mathbf{q}_{\tilde{u}}^{\diamondsuit}=\mathbf{q}_{\tilde{u}}^{\diamondsuit}\circ\mathbf{n}_{\tilde{u}}$ and the added space beam $n_{\mathrm{an},\tilde{u}}$ for UT $\tilde{u}$ cannot result in improved sum-rate performance. Thus we have

$$\sum_{k\in\mathcal{K}}\sum_{u\in\mathcal{U}} w_u\tilde{\mathcal{R}}_{u,k}\left(\mathbf{q}_u^{\mathrm{op}},\mathcal{N}_{u,1}^{\mathrm{an}}\right) = \sum_{k\in\mathcal{K}}\sum_{u\in\mathcal{U}} w_u\tilde{\mathcal{R}}_{u,k}\left(\mathbf{q}_u^{\mathrm{op}},\mathcal{N}_{u,2}^{\mathrm{an}}\right). \tag{91}$$

This completes the proof.

References

[1] W. Saad, M. Bennis, and M. Chen, "A vision of 6G wireless systems: Applications, trends, technologies, and open research problems," *IEEE Netw.*, vol. 34, no. 3, pp. 134–142, May 2020.

[2] C. D. Alwis et al., "Survey on 6G frontiers: Trends, applications, requirements, technologies and future research," *IEEE Open J. Commun. Soc.*, vol. 2, pp. 836–886, 2021.

[3] J. Wang, G. Ding, and H. Wang, "HF communications: Past, present, and future," *China Commun.*, vol. 15, no. 9, pp. 1–9, Sep. 2018.

[4] M. Uysal and M. R. Heidarpour, "Cooperative communication techniques for future-generation HF radios," *IEEE Commun. Mag.*, vol. 50, no. 10, pp. 56–63, Oct. 2012.

[5] L. Lu, G. Y. Li, A. L. Swindlehurst, A. Ashikhmin, and R. Zhang, "An overview of massive MIMO: Benefits and challenges," *IEEE J. Sel. Topics Signal Process.*, vol. 8, no. 5, pp. 742–758, Oct. 2014.

[6] M. Ke, Z. Gao, Y. Wu, X. Gao, and R. Schober, "Compressive sensing-based adaptive active user detection and channel estimation: Massive access meets massive MIMO," *IEEE Trans. Signal Process.*, vol. 68, pp. 764–779, 2020.

[7] F. Rusek et al., "Scaling up MIMO: Opportunities and challenges with very large arrays," *IEEE Signal Process. Mag.*, vol. 30, no. 1, pp. 40–60, Jan. 2013.

[8] Z. Gao et al., "Data-driven deep learning based hybrid beamforming for aerial massive MIMO-OFDM systems with implicit CSI," *IEEE J. Sel. Areas Commun.*, vol. 40, no. 10, pp. 2894–2913, Oct. 2022.

[9] Z. Gao, S. Liu, Y. Su, Z. Li, and D. Zheng, "Hybrid knowledge-data driven channel semantic acquisition and beamforming for cell-free massive MIMO," *IEEE J. Sel. Topics Signal Process.*, vol. 17, no. 5, pp. 964–979, Sep. 2023.

[10] X. Yu, A.-A. Lu, X. Gao, G. Y. Li, G. Ding, and C.-X. Wang, "HF skywave massive MIMO communication," *IEEE Trans. Wireless Commun.*, vol. 21, no. 4, pp. 2769–2785, Apr. 2022.

[11] D. Shi, L. Song, W. Zhou, X. Gao, C.-X. Wang, and G. Y. Li, "Channel acquisition for HF skywave massive MIMO-OFDM communications," *IEEE Trans. Wireless Commun.*, vol. 22, no. 6, pp. 4074–4089, Jun. 2023.

[12] D. Shi, L. Song, X. Gao, J. Wang, M. Bengtsson, and G. Y. Li, "Beam structured signal detection for HF skywave massive MIMO communications," in *Proc. IEEE 98th Veh. Technol. Conf. (VTC-Fall)*, Oct. 2023, pp. 1–6.

[13] L. Song, D. Shi, L. Gan, and X. Gao, "Slepian transform based detectors for HF skywave massive MIMO-OFDM systems," *IEEE Trans. Veh. Technol.*, vol. 72, no. 10, pp. 13106–13119, Oct. 2023.

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[14] H. Weingarten, Y. Steinberg, and S. Shamai (Shitz), "The capacity region of the Gaussian multiple-input multiple-output broadcast channel," *IEEE Trans. Inf. Theory*, vol. 52, no. 9, pp. 3936–3964, Sep. 2006.
[15] N. Fatema, G. Hua, Y. Xiang, D. Peng, and I. Natgunanathan, "Massive MIMO linear precoding: A survey," *IEEE Syst. J.*, vol. 12, no. 4, pp. 3920–3931, Dec. 2018.
[16] A. Kammoun, A. Müller, E. Björnson, and M. Debbah, "Linear precoding based on polynomial expansion: Large-scale multi-cell MIMO systems," *IEEE J. Sel. Topics Signal Process.*, vol. 8, no. 5, pp. 861–875, Oct. 2014.
[17] X. Gao, O. Edfors, F. Rusek, and F. Tufvesson, "Linear pre-coding performance in measured very-large MIMO channels," in *Proc. IEEE Veh. Technol. Conf. (VTC Fall)*, Sep. 2011, pp. 1–5.
[18] J. Hoydis, S. ten Brink, and M. Debbah, "Massive MIMO in the UL/DL of cellular networks: How many antennas do we need?," *IEEE J. Sel. Areas Commun.*, vol. 31, no. 2, pp. 160–171, Feb. 2013.
[19] C. Peel, B. Hochwald, and A. Swindlehurst, "A vector-perturbation technique for near-capacity multiantenna multiuser communication—Part I: Channel inversion and regularization," *IEEE Trans. Commun.*, vol. 53, no. 1, pp. 195–202, Jan. 2005.
[20] L. D. Nguyen, H. D. Tuan, T. Q. Duong, and H. V. Poor, "Multi-user regularized zero-forcing beamforming," *IEEE Trans. Signal Process.*, vol. 67, no. 11, pp. 2839–2853, Jun. 2019.
[21] M. Joham, K. Kusume, M. Gzara, W. Utschick, and J. Nossek, "Transmit Wiener filter for the downlink of TDDDS-CDMA systems," in *Proc. IEEE 7th Int. Symp. Spread Spectr. Techn. Appl.*, Sep. 2002, pp. 9–13.
[22] M. Sadek, A. Tarighat, and A. H. Sayed, "A leakage-based precoding scheme for downlink multi-user MIMO channels," *IEEE Trans. Wireless Commun.*, vol. 6, no. 5, pp. 1711–1721, May 2007.
[23] S. S. Christensen, R. Agarwal, E. De Carvalho, and J. M. Cioffi, "Weighted sum-rate maximization using weighted MMSE for MIMO-BC beamforming design," *IEEE Trans. Wireless Commun.*, vol. 7, no. 12, pp. 4792–4799, Dec. 2008.
[24] Q. Shi, M. Razaviyayn, Z.-Q. Luo, and C. He, "An iteratively weighted MMSE approach to distributed sum-utility maximization for a MIMO interfering broadcast channel," *IEEE Trans. Signal Process.*, vol. 59, no. 9, pp. 4331–4340, Sep. 2011.
[25] R. Sun, C. Wang, A.-A. Lu, X. Gao, and X.-G. Xia, "Precoder design for massive MIMO downlink with matrix manifold optimization," *IEEE Trans. Signal Process.*, vol. 72, pp. 1065–1080, 2024.
[26] X. Yu, X. Gao, A.-A. Lu, J. Zhang, H. Wu, and G. Y. Li, "Robust precoding for HF skywave massive MIMO," *IEEE Trans. Wireless Commun.*, vol. 22, no. 10, pp. 6691–6705, Oct. 2023.
[27] L. Song, D. Shi, L. Gan, and X. Gao, "Robust precoding for HF skywave massive MIMO with slepian transform," *IEEE Trans. Commun.*, vol. 72, no. 6, pp. 3752–3765, Jun. 2024.
[28] C. Shen and M. P. Fitz, "MIMO-OFDM beamforming for improved channel estimation," *IEEE J. Sel. Areas Commun.*, vol. 26, no. 6, pp. 948–959, Aug. 2008.
[29] M. Sandell and V. Ponnampalam, "Smooth beamforming for OFDM," *IEEE Trans. Wireless Commun.*, vol. 8, no. 3, pp. 1133–1138, Mar. 2009.
[30] E. Jeon, M. Ahn, S. Kim, W. B. Lee, and J. Kim, "Joint beamformer and beamformee design for channel smoothing in WLAN systems," in *Proc. IEEE 92nd Veh. Technol. Conf. (VTC-Fall)*, Nov. 2020, pp. 1–6.
[31] F. Jiang, Q. Li, and X. Chen, "Channel smoothing for 802.11ax beamformed MIMO-OFDM," *IEEE Commun. Lett.*, vol. 25, no. 10, pp. 3413–3417, Oct. 2021.
[32] W. Hu, F. Li, and Y. Jiang, "Phase rotations of SVD-based precoders in MIMO-OFDM for improved channel estimation," *IEEE Wireless Commun. Lett.*, vol. 10, no. 8, pp. 1805–1809, Aug. 2021.
[33] L. E. Sweeney, "Spatial properties of ionospheric radio propagation as determined with half-degree azimuthal resolution," Stanford Electron. Laboratories, Stanford, CA, USA, Tech. Rep. 155, 1970.
[34] G. A. Fabrizio, *High Frequency Over-the-Horizon Radar: Fundamental Principles, Signal Processing, and Practical Applications*. New York, NY, USA: McGraw-Hill, 2013.
[35] G. J. Frazer, "Experimental results for MIMO methods applied in over-the-horizon radar," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 32, no. 12, pp. 52–69, Dec. 2017.
[36] A. Sayeed, "Deconstructing multiantenna fading channels," *IEEE Trans. Signal Process.*, vol. 50, no. 10, pp. 2563–2579, Oct. 2002.
[37] Z. Zhang, Z. Zhao, K. Shen, D. P. Palomar, and W. Yu, "Discerning and enhancing the weighted sum-rate maximization algorithms in communications," 2023, *arXiv:2311.04546*.
[38] A. L. Moustakas, G. C. Alexandropoulos, A. Polydoros, I. Kaddas, and I. Dagres, "Impact of imperfect channel estimation in HF OFDM-MIMO communications," in *Proc. IEEE 30th Annu. Int. Symp. Pers., Indoor Mobile Radio Commun. (PIMRC)*, Istanbul, Turkey, Sep. 2019, pp. 1–6.
[39] M. Balser and W. B. Smith, "Some statistical properties of pulsed oblique HF ionospheric transmissions," *J. Res. Nat. Bur. Standards, D, Radio Propag.*, vol. 66D, no. 6, pp. 721–730, Nov. 1962.
[40] N. Maslin, *HF Communications: A Systems Approach*. London, U.K.: Pitman, 1987.
[41] Q. Hu, Y. Cai, Q. Shi, K. Xu, G. Yu, and Z. Ding, "Iterative algorithm induced deep-unfolding neural networks: Precoding design for multiuser MIMO systems," *IEEE Trans. Wireless Commun.*, vol. 20, no. 2, pp. 1394–1410, Feb. 2021.
[42] L. You, X. Gao, A. L. Swindlehurst, and W. Zhong, "Channel acquisition for massive MIMO-OFDM with adjustable phase shift pilots," *IEEE Trans. Signal Process.*, vol. 64, no. 6, pp. 1461–1476, Mar. 2016.
[43] G. Wunder, S. Stefanatos, A. Flinth, I. Roth, and G. Caire, "Low-overhead hierarchically-sparse channel estimation for multiuser wide-band massive MIMO," *IEEE Trans. Wireless Commun.*, vol. 18, no. 4, pp. 2186–2199, Apr. 2019.
[44] *Proplab-Pro Version 3*. Solar Terrestrial Dispatch. Accessed: Mar. 2021. [Online]. Available: https://shop.spacew.com/index.php/product/proplab-pro-hf-radio-propagation-laboratory/
[45] R. Penrose, "A generalized inverse for matrices," *Proc. Cambridge Philos. Soc.*, vol. 51, no. 3, pp. 406–413, Jul. 1955.



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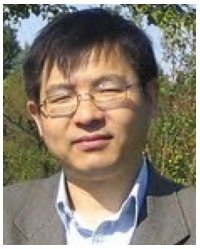
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